# AFE BABALOLA UNIVERSITY

# ADO-EKITI, NIGERIA.

## COLLEGE OF ENGINEERING

## DEPARTMENT OF ELECTRICAL ELECTRONICS AND COMPUTER ENGINEERING

## UNDERGRADUATE HANDBOOK

B.Eng. Electrical Electronics Engineering

**2026 – 2030**

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## A BRIEF HISTORY OF THE UNIVERSITY

Afe Babalola University is the brainchild of Chief Afe Babalola, B.Sc. (Econs) Lond; LL.B (Hons) Lond; BL; FFPA; FNIALS, FCI Arb; LL.D; Senior Advocate of Nigeria (SAN); Doctor of Letters (LL.D); Officer of the Order of the Federal Republic (OFR), and Commander of the Order of the Niger (CON), the immediate past Pro-Chancellor and Chairman of the Governing Council of the University of Lagos, Nigeria, and the immediate past Chairman of the Committee of Pro-Chancellors of Nigerian Universities.

It is a product of a mind disturbed by the incurable problems afflicting government universities and the quality of graduates being produced each year. As a Pro-chancellor and Chairman of Pro-chancellors, he found that these Universities problems were due to lack of commitment on behalf of the staff and want of emotional attachment to their various institutions as in the case of a private or personal establishment. It is common knowledge that workers in public institutions often display lackadaisical attitude towards government work with the justification that it is not their father's business.

The Promoter believes that a time has come when private individuals should stand up to the challenge of providing affordable and qualitative tertiary institutions as it is done in other nations like the ivy league of Universities in the United States and other leading institutions in Europe. These are classic examples of how and why private universities are more successful than government universities evidenced by the quality of graduates they produce and their performances in the competitive world field. For example, Japan which has the largest number of private Universities is indisputably a leader in the field of technology which goes to show that private institutions offer brighter prospects of qualitative tertiary education.

This is because the proponents of these great private universities have dreams, ideas, and objectives about the type of university they want. They established code of conduct and disciplinary measures which are pursued religiously and rigorously.

**Location**

The University is located on a 130 hectares land at an altitude of over 1500 ft above sea level along Ado-Ekiti - Ijan Road, directly opposite the Federal Polytechnic, Ado-Ekiti in Ekiti State, which ipso facto provides cool and ideal climate for learning and sport activities.

**Vision**

The vision of the University is

> "***to be a world class educational centre of excellence in academics, character, sports and vocational development***"

The vision is guided by the need to produce professionals who are sound and agile. Through its academic and professional programmes, graduates of the University will emerge as people with professional skills and become business owners thus assuring each of them a job after graduation.

The Promoter dreams of producing leaders and achievers who would be self-reliant, kind and generous, considerate and sportsman like. The Promoter's guiding principle that ***he prays most who works hardest*** shall be taught, applied and practiced religiously. Students shall be made to believe in the Promoter's golden rule that no matter one's background, nothing is impossible; and with hard work, one can make it to the top.

In form and structure, Afe Babalola University shall be a citadel for the promotion of an authentic learning environment. It shall put intellectualism at the centre of its tradition.

The temperament of universality and excellence shall be imbibed and shall be evident at every level of academic decisions-making in all colleges and departments of Afe Babalola University.

This vision is in line with the policy of the Federal Government which encourages public-private partnership. As he did in University of Lagos, the Promoter would deploy his vast contacts in business and professional circles to muster support for the fulfillment of the vision.

**Mission**

> *Afe Babalola University is a result-oriented institution for producing highly skilled and socially relevant graduates capable of applying scientific knowledge for the resolution of social problems. The University is equally committed to transforming students into expert thinkers, innovative managers and resourceful technocrats in all fields of learning.*

**The Philosophy**

The Promoter of the University strongly believes in the production of graduates who appreciate the value of the dignity of labor and who are problem solvers, self-reliant, employers of labor, highly productive and globally relevant in every sphere of human endeavor.

**Goals and Objectives**

To achieve its mission and fulfill its purposes, Afe Babalola University has set forth specific goals and objectives. The goals describe the results desired from the University's Educational Programmes, and the objectives set forth general guidelines to be followed on the road to reaching those goals.

**Goals**

As a unique University, it seeks to achieve the following goals:

i. Encourage a balance between academic excellence, professional ethics, and practical competence among students;
ii. Encourage daily consistency and personal integrity among the students. To pursue academic excellence through outstanding teaching and the promotion of research and community service on the national and international level. To create and promote interdisciplinary and international scholarship and education, while sustaining foundational disciplines;
iii. Define and lead emerging disciplines and intellectual paradigms that create knowledge and to promote the translation of that knowledge to the benefit of society;

iv. Develop new learning opportunities that build on these paradigms and create student engagement in the context of a community of scholars; and
v. Foster effective concern for environmental well-being, justice, peace and compassion and commitment to their promotion in the contemporary world.

**Objectives**

In order to achieve these goals, we pursue the following objectives:

(a) Establish in Ado Ekiti a world class institution of higher learning which combines the acquisition of knowledge with its practical application in training graduates of high integrity and of international recognition;

(b) Provide an environment of international standards for learning, research, acquisition of specialized skills and comprehensive training of students empowering them to apply knowledge and skill to the benefit of society;

(c) Complement the efforts of Government in bridging the yawning unemployment gap by creating employment opportunities for Nigerians.

(d) Develop and continuously up-date a modern library including a digital library to encourage research, publication and to promote a "community of learning" atmosphere, global interaction as well as promote scholarship and conduct research in all fields of human endeavor to advance learning and provide solutions to developmental problems for the benefit of society and mankind;

(e) Develop and offer specialized, flexible and innovative programmes for specialized and non-specialized undergraduate and post graduate degrees and non-degree programmes, thus creating a balance between academic excellence, professional ethics and practical competence among a critical mass of human capital who are imbued with a high sense of responsibility, individual initiative as well as with capacity for analytical, creative and practical thinking;

(f) Promote effective communication among the academic community, students and the public in order to produce specialists competent to bring about research aimed at creative inventions and practical contribution towards the advancement of knowledge, technological development and for the creation of wealth for Nigeria.

(g) Establish a centre for entrepreneurial studies to stimulate job creation abilities in students from the onset of their studies;

(h) Establish a Creative and Lifelong Education Centre for the purpose of facilitating non-degree and vocational skills programmes;

(i) Encourage and promote the advancement of learning and hold out to all persons without distinction of race, creed, sex, religious or political conviction the opportunity to acquire university education; and

(j) Undertake other activities appropriate for teaching, research and community service as expected of a university of high standard.

**Strategies for Achieving Stated Objectives**

The major University-wide strategies to be adopted in achieving the objectives of Afe Babalola University are premised on a platform of these basic characteristics:

a. Assistance by way of scholarships and job opportunities on campus to brilliant but indigent students;
b. Excellence in educational programmes that is affordable to students;
c. Commitment to attracting and retaining world-class teaching and non-teaching staff, along with competitive employee compensation and support for success;
d. Institutional agility and resource flexibility in order to capture emerging opportunities, and to be readily and creatively responsible to the needs of constituent stakeholders at the state, national, and international levels.
e. Undergraduate and post-graduate programmes in various disciplines that meet the students' need for learning and the societal need for solutions to arts, science and technological challenges. The University is accessible to diverse makers, and offers curricula designed to serve their needs.
f. Courses are broad-ranged and integrated to facilitate the required creativity and productivity of students.
g. The University emphasizes student-centered learning environments engaging academic variety and student diversity, with the goals of providing a holistic and collaborative approach to learning; learning through discovery, experience, and innovation; and effective use of information.
h. The University has developed innovative means of providing financial assistance to deserving indigent students, with due regard to competitive recruitment of qualified students, cost of education and student need, enrolment characteristics, scholastic achievement, and educational enrichment.
i. The University has developed information technology and widespread access as a catalyst in the execution of other strategies, especially in the creation of new mechanisms for effective engagement of service.
j. The University has created and expanded academic facilities lecture rooms, e-learning platforms, ICT center, library, laboratories, and studios – to support innovative teaching and learning.
k. The University provides learning experiences for students through hands-on community service opportunities and internship opportunities that complement their career objectives.
l. The University publicizes and promotes institutional achievements and staff and student accomplishments among the constituent stakeholders and enhance national and international visibility.
m. There will be formal assessment of student outcomes at the University level at least every other year and, more frequently, there will be assessment of the outcomes and effectiveness of academic programmes, courses, and targeted initiatives in the interest of continuous improvement.

n. All students will take compulsory courses that cut across all programmes being undertaken. They include courses such as Phonetics and Use of English, Communication Skills, Logic, Creative Thinking, African Culture and Ethics, Entrepreneurship as well as basic education in practical agriculture.
o. The University operates a non-discriminatory students' admissions policy on the basis of gender, tribe, religion, political leanings or physical disability. Only candidates that are highly qualified and able to benefit from good university education will however stand a good chance of being admitted.
p. Admission of students is through the UTME including a selection process that will from time to time be determined by the University.
q. The method of instruction is face-to-face interaction – a normal 17 weeks semester for roper educational impartation of knowledge. The use of modern communication facilities available in imparting knowledge to students will be employed. Practical and tutorials are also designed and students are given opportunities to think out projects and work under supervision. On-line approach to student-teacher communication is in place to facilitate student-teacher interactions. To this end every staff and student will own a computer for their use.
r. There will be continuous assessment as part of the grading system to ensure that students are up to date in their studies.
s. The University undertakes a periodic curriculum review that takes into account feedback from organizations and employs of labour and NUC Minimum Academic Standards.
t. All students will have access to welfare, career planning and placement services, and accommodation in well-equipped and functional halls of residence.
u. Opportunities will be provided for staff and facilities evaluation exclusively by students.

## FORWARD FROM HEAD OF DEPARTMENT

Welcome to the B.Eng. Electrical and Electronics Engineering programme at Afe Babalola University, Ado-Ekiti (ABUAD). Since our inception in September 2010, the programme has established itself as a world-class center of excellence, committed to fostering academic innovation, ethical leadership, and holistic development in character and research. Our vision is to produce highly competent engineers equipped with the technical expertise and professional ethics necessary to address global engineering challenges and apply scientific knowledge to the resolution of complex technical and social problems.

The programme's philosophy is rooted in producing graduates who appreciate the value of the dignity of labor and emerge as self-reliant problem solvers, employers of labor, and globally relevant expert thinkers. To support this, our curriculum follows a rigorous Outcome-Based Education (OBE) framework, integrating twelve critical Programme Outcomes that align with COREN and national standards. We offer specialized training through "signature courses" as part of our 30% local content specialization, supported by a state-of-the-art learning environment comprising nine specialized laboratories, including the National Instrument Academy and Data Acquisition Laboratory.

We place immense value on bridging the gap between theory and industrial practice through mandatory hands-on training and extensive industrial exposure, such as our 24-week SIWES placement. Through strategic partnerships with industry leaders and the integration of professional IT certifications like CCNA, Machine Learning, and Python programming into our curriculum, we ensure our students acquire the vital technical capabilities and workplace ethics required for the 21st-century workforce. Guided by our active Industry Advisory Board, we are dedicated to nurturing innovation and principled leadership, empowering our graduates to excel in their professional careers and contribute to the advancement of the engineering profession globally.

**Engr. Dr. Adedayo Olukayode Ojo**
*Ph.D. (ABUAD, 2024), M.Eng. (UPM, 2014), B.Eng. (LAUTECH, 2008), COREN, MIEEE, MNSE*
***Ag. Head, Department of Electrical, Information and Computer Engineering***

## B. ENG. ELECTRICAL AND ELECTRONICS ENGINEERING

### The Programme

The field of electrical and electronics engineering could be divided into two broad categories: light current electrical engineering and heavy current electrical engineering. The light current electrical engineering includes electronics, radio, communications, instrumentation, measurements, control, and materials science and devices technology. The category of heavy current electrical engineering covers electrical power systems, high voltage technology, and electromechanical energy conversion including machines and power electronics. All these special branches are widely utilized in electrical engineering industries, corporations and other government establishments. The undergraduate studies programmes in the Department are aimed at the training of high-level manpower in the specialized areas of electrical engineering for national development. The courses are designed to train the students in the latest developments and modern technologies in the solution of electronics and electrical engineering problems facing the nation.

### Programme Vision

> *To be a **world-class center of excellence** in Electrical/Electronics Engineering education, fostering academic **innovation**, ethical leadership, and holistic development in **character**, **research**, and **vocational** competence.*

### Programme Mission

> *The Electrical/Electronics Engineering programme is a **result-oriented** programme **for producing highly skilled and socially relevant Electrical/Electronics Engineering graduates capable of applying scientific knowledge for the resolution of social problems**, and **equally committed to transforming students into expert thinkers, innovative managers, and resourceful technocrats in all fields** of Electrical/Electronics Engineering.*

### *Philosophy*

The programme strongly believes in the production of graduates who appreciate the value of the dignity of labor and who are problem solvers, self-reliant, employers of labor, highly productive and globally relevant in every sphere of human endeavor.

***Programme Educational Objectives (PEOs)***

**PEO1** *To produce highly skilled and socially relevant electrical/electronics engineering graduates who achieve excellence in academics and contribute to professional, industrial, and academic practice.*

**PEO2** *To produce proficient and self-reliant resourceful technocrats who represent a centre of excellence in addressing contemporary challenges through the application of scientific knowledge.*

**PEO3** *To produce electrical/electronics engineering graduates committed to continuous vocational development and lifelong learning, ensuring they remain expert thinkers who are socially relevant in a rapidly evolving engineering environment.*

**PEO4** *To transform students into expert thinkers and innovative managers who provide result-oriented and resourceful solutions for the resolution of social and real-world problems across all fields of electrical/electronics engineering.*

**PEO5** *To produce electrical/electronics engineering graduates of high character who uphold integrity and ethical standards, acquiring and applying scientific knowledge responsibly in line with sustainable development principles.*

***Programme Outcomes (POs)***

| | | |
|---|---|---|
| **PO1** | **Engineering Knowledge** | Apply knowledge of mathematics, science, engineering fundamentals and electrical/electonics engineering to the solution of developmental and complex engineering problems. |
| **PO2** | **Problem Analysis** | Identify, formulate, research literature, and analyze developmental and complex computer engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences |
| **PO3** | **Design/Development of Solutions** | Proffer solutions for developmental or complex electrical/electronics engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations |
| **PO4** | **Investigation** | Conduct investigation into developmental or complex electrical/electonics engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to |

| | | |
|---|---|---|
| | | provide valid conclusions |
| **PO5** | **Modern Tool Usage** | Create, select, and apply appropriate techniques, resources and modern engineering and ICT tools, including prediction, modeling, and optimization to developmental and complex computer engineering activities, with an understanding of the limitations |
| **PO6** | **The Engineer and Society** | Apply reasoning informed by contextual knowledge including Humanities and Social Sciences to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional electrical/electonics engineering practice |
| **PO7** | **Environment and Sustainability** | Understand the impact of professional electrical/electonics engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development |
| **PO8** | **Ethics** | Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice, including adherence to the COREN Engineers Code of Conducts |
| **PO9** | **Individual and Teamwork** | Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings |
| **PO10** | **Communication** | Communicate effectively on developmental or complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions |
| **PO11** | **Project Management** | Demonstrate knowledge and understanding of electrical/electonics engineering, management and financial principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments |
| **PO12** | **Lifelong Learning** | Recognize the need for and have the preparations and ability to engage in independent and lifelong learning in the broadest context of technological and social changes. |

*Admission Requirements for all Engineering Programmes*

**ADMISSION REQUIREMENTS**

(a) **O' Level Subjects**

Five O' Level credits to include Mathematics, English Language, Physics, Chemistry and any other subject from Biology, Agriculture, Technical Drawing, Geography.

(b) **UTME Subjects**

English Language, Mathematics, Physics and Chemistry. Expected duration for UTME candidates shall be 5 years.

(c) **Direct Entry**

(i) At least two GCE Advanced Level passes which must include Economics, and any other subject from Government, Geography, History, Sociology, Psychology, Mathematics, and Further Mathematics.

(ii) IJMB/JUPEB/NCE/ND or its approved equivalent at an acceptable grade level.

(iii) In addition, candidates must satisfy the O' Level requirements as stated in (a) above.

| **Admission requirements** | | **Remarks** |
|---|---|---|
| **SSCE** | **Direct Entry** | |
| Five credit passes in not more than two sittings in English language, Mathematics, Physics, Chemistry, and any other Science subjects in WASSCE, GCE or NECO<br><br>It is also desirable for candidates to have further Mathematics and Technical Drawing at credit levels. Such candidates shall have added advantage. | Advanced level passes in Mathematics, Chemistry and Physics in addition to UTME requirement.<br><br>Candidates with National Diploma at Upper credit Level may be considered. | **UTME subjects**<br><br>The joint Matriculation Examination must be in use of English language, Mathematics, Physics and Chemistry. |

**Inter-University Transfer**:

Candidates from other accredited institutions can enrol in the Programme. The university rule stipulates that every candidate must complete a minimum of two academic terms at the institution to receive the university degree. Those possessing a Higher National Diploma in a related area may qualify for entry into the 300 level. Each candidate is assessed according to their own qualifications and achievements.

*Registration*

Students are strongly advised to carefully study and follow the prevailing registration procedures.

**LATE REGISTRATION**: The University frowns at this and as a result, students should avoid it. Any defaulter shall be sanctioned accordingly.

***Examination, Grading, Probation and Withdrawal***

**EXAMINATION**

**(i) Continuous Assessment and Attendance**

There shall be continuous assessment in each course which shall carry a mark of thirty percent (30%) and ten percent (10%) attendance.

**(ii) Semester Examination**

There shall be a written examination of 2-3 hours duration in each course at the end of the semester in which the course is concluded. Examination shall carry a minimum of sixty percent (60%).

**GRADING**

The grading system used for examination classification is given as follows:

| MARKS | CREDIT | POINTS |
|---|---|---|
| 70 - 100 | A | 5 |
| 60 - 69 | B | 4 |
| 50 - 59 | C | 3 |
| 45 - 49 | D | 2 |
| 0 - 44 | F | 0 |

**CLASS OF THE DEGREE**

The list of successful candidates for the degree examination is normally published by the Registrar. Degrees are usually classified as First Class Honors, Second Class Honors (Upper Division), Second

Class Honors (Lower Division), Third Class Honors. The University will not award a Pass Degree. The class of the degree is awarded on the basis of Cumulative Grade Point Average (CGPA) as indicated below:

| CGPA | CLASS OF DEGREE |
|---|---|
| 4.50 – 5.00 | First Class |
| 3.50 – 4.49 | Second Class (Upper Division) |
| 2.50 – 3.49 | Second class (Lower Division) |
| 1.50 – 2.49 | Third Class |

Any student who has a Cumulative Grade Point Average (CGPA) of less than 1.50 at the end of the first semester of first year shall be placed on College warning. If at the end of the second semester, such student fails to obtain a CGPA of 1.50, he/she shall be placed on University warning. At the end of the third semester, if CGPA is still less than 1.50, he/she shall be placed on probation. If the CGPA fails to exceed 1.50 at the end of the fourth semester, he/she shall be advised to withdraw from the programme.

**MINIMUM UNITS AND DURATION REQUIRED FOR THE AWARD OF B. ENG. DEGREE**

Minimum duration of Engineering programme is five academic sessions for candidates who enter with Senior secondary School Certificate or GCE 'O' Level qualifications. Candidates who come in by Direct Entry will spend a minimum of four academic sessions provided that they satisfy all the other University requirements.

For a candidate to qualify for the award of the Bachelor of Engineering (B. Eng.) Degree, he/she must have passed the minimum total units in each programme. No student is expected to have less than 1.50 as his/her Grade point Average (GPA). In addition, the student must have passed all the GST courses. The Cumulative Grade Point Average (CGPA) shall be used for the determination of the class of degree.

**OTHER REQUIREMENTS FOR THE AWARD OF B. ENG. DEGREE**

**Grade Point Average and Cumulative Grade Point Average**

Grading of courses is done by a combination of percentage marks and letter grades translated into a graduated system of Grade Points Equivalents (GPE). The academic performance of a student is measured by the Cumulative Grade Point average (CGPA). The GPA for a semester is obtained by multiplying the appropriate grade point by the total number of units of the courses and the products are added for all the courses. The sum is divided by the total number of units of the courses taken for the semester. The CGPA on the other hand is obtained by adding up the weighted grade points for all the semesters. The sum is divided by the total units offered for all semesters.

Each course is graded out of maximum of 100 marks and assigned appropriate Grade Point Equivalent as in the table below. Each course has its credit assigned to it. For example, ENG 381 has 3 Units, while CHE 311 has 2 Units and CHE 421 has 2 Units.

Grade Point Average Calculations

| Courses | (i) | (ii) | (iii) | (iv) | (v) |
|---|---|---|---|---|---|
| | Assigned Credit Units | Score earned in % | Points earned for scores in (ii) | Grade Points (GP) | Grade Point Average (GPA) |
| ENG 381 | 3 | 70 | 5 | $3 \times 5 = 15$ | $GPA = \frac{29}{7} = 4.14$ |
| CHE 311 | 2 | 58 | 3 | $2 \times 3 = 6$ | |
| CHE 421 | 2 | 64 | 4 | $2 \times 4 = 8$ | |
| Total Credits registered | 7 | | | Summation of (iv) is 29 | |

**MINIMUM AND MAXIMUM LOADS**

The minimum workload for a student is 15 units per semester, while the maximum is 24 units.

**COURSES OFFERED**

**100 Level**

All common courses at the 100 level are taken at the college of science to boost their knowledge of the basic sciences required for any engineering programme.

**200 Level**

At the 200 level, all Engineering students undertake common courses at the college of Engineering which are required to develop them in accordance with the requirements of the Engineering Council for the training of Engineers in Nigeria.

**300 Level to 500 Level**

The 300 Level initiates the beginning of the respective programme hence all students of each department gather to take courses planned for their respective programmes with a few relevant general Engineering courses offered at the College. Whereas at 400 level and 500 level, only the requisite courses for individual programmes are taken. The detailed course structure is as presented in other sections.

## INDUSTRIAL TRAINING AND ASSESSMENT

Engineering education is incomplete without industrial attachment being part of the degree programme as recommended by the NUC. This is because it exposes students to the real or practical environment of the profession. At ABUAD the training is carried out as follows:

**SWES I encoded ENG 290:** This is carried out by the college as an in-house workshop practical training in the various Engineering fields within the College and around the campus. It occurs for about two weeks after the end of the 200 level Session Examinations. During the period, the students are exposed to workshop practices in the mechanical, machine, sheet metal, automobile, welding, carpentry, civil and electrical engineering workshops.

**SWES II encoded ENG 390 (for 300 Level):** The attachment takes place at the end of 300 level session examinations during the long vacation. Students are attached to various relevant establishments or industries outside the university for about three months. During this period, academic engineering staff members pay visits to the students and assess as well as encourage them on the essence of the training.

**SIWES III encoded ENG 490 (for 400 Level):** That of the 400 Level starts at the end of the first semester examinations and last for 24 weeks. On completion of the attachment programme the student is assessed by the engineering staff who visited the students on training, together with grading of the logbooks and a written report on each student experiences on the establishment.

## CONDUCT OF EXAMINATIONS:

***Examination and Evaluation***

Students will be examined and evaluated in all courses registered for. Basic theoretical courses will be evaluated through continuous assessments and end of semester examinations. It is mandatory for students to perform satisfactorily in both the continuous assessment and examinations to progress in the student's choice of study. Various methods of evaluation to be adopted are as described below:

i. ***Continuous Assessment:***

In the college of Engineering, assessment of students' achievements would be based on:

(i) Examinations
(ii) Laboratory reports
(iii) Planning, conduct and reporting of project work
(iv) Oral presentations and problem-solving exercises
(v) Assignment
(vi) Group project work
(vii) Reports of Industrial Training programme.

Continuous assessment shall be done through assignments, tests and practical exercises.

(a) Scores from continuous assessment shall normally constitute 30% of the final marks for courses which are primarily theoretical.

(b) For courses which are partly practical and partly theoretical, scores from continuous assessment shall constitute 50% of the final marks.

(c) For courses that are entirely practical, continuous assessment shall be based on a student's practical work or reports and shall constitute 100% of the final marks.

- ***Group Work:*** Here students are given liberty to discuss with fellow group members for individual conclusions to be drawn.
- ***Assignments:*** These will be graded by lecturers. The University frowns on any student who 'steals' ideas from fellow students or plagiarizes. It is expected that all assignments will be written in English Language. Completion of assignments by students and marking by staff will be tied to time, as deadlines will be given for both. Marked scripts will be returned to students on specified dates to enable students read the lecturer's comments and evaluation. The University will orientate all appointed lecturers with the principles of fairness, diligence and conscientious service. A process of seeking redress by aggrieved students will be instituted.
- ***Examination:*** This is used for students' overall performance during the semester. There will be a mid-semester test and an end-of-semester examination in all the courses offered. Students will be expected to make themselves available for these tests and examinations. They will be examined in all courses offered based on the lectures given by the lecturers, recommended texts, assignment and research work during the semester. Each College and Department will make the time table available at least two weeks to the commencement of the examination. Use of mobile phones and other communication equipment will not be allowed in the examination hall as they will be regarded as tools for examination malpractices. Both the examination and continuous assessments will be used for final grading. Examination carries 60%.
- ***Class Attendance:*** Every student must have at least 90% class attendance at the end of each semester. Any student who falls below this requirement will not be allowed to sit for the end-of-semester examination except as a result of unforeseen circumstances which the University is privy to.
- ***Project****:* A student is expected to undergo an independent but supervised project before he/she completes his degree programme. The project will carry six compulsory units in all undergraduate programmes. No student will be allowed to graduate without evidence of completion of his/her independent research work.

**Discipline:**

A student may be expelled on the basis of established case of examination misconduct or any actions likely to cause breach of peace. Such actions include cultism, vandalism, or gross misconduct as may be determined by the laws of the University. Such a student will however have opportunity to defend himself before the Students' Disciplinary Committee. Once a student is expelled from the University, he has no opportunity of being re-admitted to the University again.

**Examination Conduct**

- All University Examinations shall be held in halls, lecture rooms/theatres, and or laboratories approved by the University.
- Students shall be seated far apart enough to prevent cheating, and all students shall be visible to the invigilators. Students' seats shall be arranged to make it possible for invigilators to reach them with ease.

**Attendance:** Attendance of students sitting for each examination must be taken before the start of the examination.

**Handling of Answer Booklets**

Answer booklets shall be treated as security materials, and it shall be an offence for anyone to put them to other use different from what they are meant for. Students found with unauthorized possession of Answer Booklets shall be liable to disciplinary action.

**Eligibility for Examination**

All Students who are duly registered for courses in a given semester are eligible to sit for examinations in those courses except students in the following categories:

i. Students who have not paid up their school fees and other levies;
ii. Students who fail to attend up to 90% of practical/lecture hours;
iii. Students on suspension for one reason or another.

**Absence from Examination**

1. Students must present themselves at University examinations for which they have registered under the regulations. Students who fail to do so for reasons other than illness or accident shall normally be deemed to have failed the examination and would be scored zero. Mis-reading of Time-Table shall not be accepted as a satisfactory explanation for absence.
2. Any student who, on account of ill-health, is absent from a University Examination may be permitted by the Senate after considering a report from the appropriate College Board, to present himself for such examination without penalty, at a future date provided that:
   (a) The illness has been reported to the Registrar; and
   (b) The student has been examined by the University Medical Officer or a Medical Report submitted to the Registrar within a reasonable period, not exceeding forty-eight hours of the absence, under this regulation.

**Medical attention**

University Medical Officers shall normally be on call for the purpose of attending to students for the whole period of examination.

**Typing or Dictation of Answers**

Where a student duly registered for any University Examination and for special medical reasons, desires to use computer or dictate his answer, the Deputy Registrar (Academic) shall, after consultation with the Director of Medical Services, arrange for the typing or dictation of his answer under supervision, at the student's expense.

**Instructions to Candidates on Examinations Specific**

Students shall:

(a) use or consult, during an examination, only such books, papers, instruments or other materials or aids as are specifically permitted or provided by the Department in which the examination is being held;
(b) not introduce or attempt to introduce into the examination room/hall hand bags, books, notes, jotters, instruments or other materials or aids that are forbidden;
(c) not enter any examination room with any inscription on any part of the body, e.g. palm,

arm, thigh, etc. especially if such inscription bears any relevance to the examination;

(d) not pass or attempt to pass any information from one person to another during examination;

(e) neither sit in collusion with any other student(s) or person(s) nor copy, nor attempt to copy from another student, nor engage in any similar act;

(f) not disturb nor distract any other student (s) during the examination;

(g) not employ/engage other people to write any University examination on their behalf;

(h) obey invigilators' instructions to ensure the smooth conduct of the examination.

2. **General Rules on Examinations**

Students shall:

(a) always ensure that they acquaint themselves with the examination regulations and instructions to students;

(b) attend/report punctually at the time scheduled for their examination papers. No student shall be allowed to enter an examination room/hall earlier than ten (10) minutes before the commencement of the examination. Any student who arrives ten (10) minutes after the examination has started, shall be admitted only at the discretion of the Chief Invigilator. Such student(s) shall not be given extra time. Misreading of the examination Time-Table is not acceptable excuse for lateness to the examination hall;

(c) bring with them to the examination hall their writing materials and other materials which are permitted under these regulations;

(d) keep strictly to the sitting arrangement in the examination room to avoid confusion. Chairs arranged in the hall for examination purposes should not, under any circumstances, be removed. Students wishing to do some revision before the commencement of the Examination shall do so outside the examination hall;

(e) not be allowed to leave the examination hall after question papers have been distributed, without being accompanied by a member of staff. Also, no student shall be allowed to leave the examination hall within the first thirty (30) minutes of the examination and thirty (30) minutes to the end of the examination;

(f) use only answer booklets and supplementary answer sheets that carry the College Stamp and date of the examination. Answer booklets are security materials which shall be stamped by each College for each day's examinations. It shall be an offence for any student to take answer booklet(s) out of the examination hall or put such to use thereafter other than for University examination;

(g) write their matriculation numbers clearly on the answer booklet immediately they are seated, and before the commencement of the examination;

(h) place on the table, their University Identity (ID) Card for the invigilators' inspection at any time during the examination;

(i) observe silence in the examination hall. The only permissible way of attracting the attention of the invigilator is by show of the hands. While the examination is in progress, communication of any kind between students is strictly prohibited. Any student found to be giving or receiving irregular assistance shall be guilty of misconduct for which he/she may be penalized;

(j) not speak in the examination hall as this is strictly prohibited;

(k) not use scrap paper(s). All rough works should be done in the answer booklet and crossed neatly through,

(1) not be allowed to submit their answer scripts in the first thirty (30) minutes and last 30 (thirty) minutes of any examination. Students shall remain seated at the end of the examination while invigilators go from row to row to collect answer scripts. It is the responsibility of students to ensure that their scripts are collected by the invigilators before they leave the examination hall;

(m) satisfy themselves before handling in their scripts at the end of the examination, that they have inserted at the appropriate places, their matriculation numbers and the number of questions answered. Except for the question papers and any other materials they may have legitimately brought with them, students shall not be allowed to remove or mutilate any paper or materials supplied by the University.

(n) observe the following conditions under which calculator shall be used:

i. The use of electronic calculators, except programmable ones, may be permitted.
ii. The calculators must be small (hand-held) and battery/solar operated.
iii. No student should borrow another student's calculator during examination; this practice shall be construed as giving or receiving irregular assistance during the examination and shall constitute examination misconduct.
iv. Instruction manuals are prohibited (as these often contain useful mathematical formulae and methods). Calculator packets and containers are also prohibited in the examination halls.
v. The calculator should be switched off on entry into the examination room. Only one calculator per student is allowed.

**Penalties for Examination Misconduct**

Any student established to have cheated or aided and abetted cheating in any examination or found guilty of being in possession of incriminating materials at the examination, or involved in any other serious examination misconduct before, during or after the examination, including impersonation, shall either be rusticated for a specified period of time or expelled from the university depending on the gravity of the offence.

**Procedure for Investigating Alleged Examination Misconduct:**

1. Whenever a student is caught for any examination offence, the case shall be reported to the invigilator/supervisor in the hall immediately.
2. The invigilator shall fill in the necessary form reporting the case of examination misconduct and the student shall be made to write a statement on his or her involvement. The student shall be made to continue with the examination.
3. The invigilator/supervisor shall then report formerly to the Head of the Department of the affected student(s) by submitting all the reports.
4. The Provost will then set up a committee of three lecturers chaired by a lecturer whose rank is not lower than that of a Senior Lecturer to carry out a preliminary investigation of the alleged

examination misconduct at the College level. Members of the committee shall be from departments other than that of the student(s) affected.

5. The report of the College preliminary investigation carried out by the committee must be ready within two weeks, and shall be forwarded to the Vice-Chancellor by the Provost of the affected College.
6. The Vice-Chancellor on receiving the report, shall send it to the Students' Disciplinary Committee for determination and recommendation to him.
7. The Students' Affairs Unit, on receiving the report of the preliminary investigation, will frame appropriate charges against the student. The student will be asked to defend himself/herself against the charges preferred against him/her in writing, submitting a written statement of same. The student will then be invited to appear before the SDC to defend himself/herself verbally.
8 The Students' Disciplinary Committee shall read the charges preferred against the offender to him or her, and allow him or her to defend himself or herself in the light of his/her statements which he/she had earlier submitted.
9 The report and recommendations of the Students' Disciplinary Committee shall be forwarded to the Vice-Chancellor for consideration and approval.
10 Students may appeal against the decision of the SDC to Council within 14 days.

**Examination Misconduct and Student Discipline**

1. Any action which prejudices the integrity of the university examination shall be considered to be an academic misconduct and shall be punishable by appropriate disciplinary action
2. With respect to all proceedings under the University's rules and regulations in relation to academic discipline, each student shall be presumed innocent until the contrary is established.

**Academic Misconduct**

Without limiting the generality of the foregoing, academic misconduct shall be deemed to include, but shall not be limited to, the following:

1. Cheating in examinations, assignments, term papers, reports, projects or any other tests which are to be used in judging the students' performance in a course or programme of study, or on special tests which the University may offer. Cheating includes copying from another student's work or allowing another student to copy from one's own work, consultation with any unauthorized person during an examination or test and use of unauthorized aids;
2. Impersonating another student or entering into an agreement with another person to be impersonated for purpose of taking examinations or tests, or carrying out laboratory or other assignments;
3. Plagiarism, that is, the act of presenting the ideas or words of another as one's own. The use of another people's ideas or words must be properly acknowledged and referenced. This applies to all written materials such as essays, laboratory reports, term papers, design and other projects, statistical data, computer programmes and research results. The use of such materials without due acknowledgement, however, is contrary to accepted norms of academic behavior;
4. Obtaining by theft or other improper means of examination papers, tests, or any other materials

from other students;

5. Falsifying academic records or submitting false credentials for the purpose of gaining admission to a University examination or any other academic purposes;
6. Submitting a false medical or other certificate or obtaining such certificates under false pretence for examination or any academic purpose whatever;
7. Submitting an essay, report or assignment to satisfy some or all the requirements of a course when that essay, report or assignment has been previously submitted or is currently being submitted for another course without the express permission of the lecturer(s) involved;
8. Behaving in a manner which infringes in any unreasonable way on the right of the other students to engage in their scholarly work. This shall include the disruption of classes and examinations and the harassment or intimidation of the student or staff;
9. Behaving in a laboratory in a manner which is in serious or repeated violation of safety regulations and thereby creating a situation that constitutes a threat to the safety of individuals or other occupants of the laboratory.

**Disciplinary Action**

The SDC may recommend one or more of the following disciplinary actions:

1. Reprimand:- This shall be in the nature of a warning to the student that his/her behaviour has been unacceptable to the University and that any further behaviour of this sort shall lead to stiffer penalty;
2. Disciplinary Probation:- A student may be placed on disciplinary probation by the Provost for the balance of the period of registration at the University in the Degree Programme in which the student was registered at the time of the misconduct. A note of the effect that the student is on probation will appear in the student's file. The effect of being placed on disciplinary probation is that any further misconduct shall lead to a stiffer penalty;
3. Suspension: A student may be suspended from the University for a period which shall not exceed one academic session. While suspended, a student may not register in the University and shall lose the right to attend lectures, write examinations, and receive remuneration from University source. Notice of a suspension shall be placed in the student's file and will appear on any transcript issued during the period of suspension. Upon completion of the period of suspension, the student will be eligible for reinstatement to full academic status and the notice of suspension will be removed from the student's permanent record;
4. Expulsion: A student may be expelled from the University by the Vice-Chancellor following the recommendation of the students' disciplinary committee Expulsion shall be permanent. A statement of such action shall be part of the student's permanent academic record and shall appear on all subsequent transcripts.

**The Responsibility of the University**

1. It is hereby declared that while this policy is concerned with the nature and procedure of handling academic misconduct committed by students, the University also acknowledges certain fundamental responsibilities pertaining to such matters. In particular, the University has a responsibility to provide adequate facilities and invigilation for all examinations. Students who are required to write examinations under crowded or otherwise inadequate conditions, may be placed in a position where cheating is very difficult to avoid.

2. The University also has a responsibility to ensure that teaching staff uphold the principles of the University and the integrity of the scholarly activity in which they are engaged.

**STAFFING**

**Academic Staff**

| S/N | Name | COREN # | Rank | Date of first appointment | Details of Qualifications | | | Specialization | Experience Teaching (Total) Years | Dedicated /Shared | Credit Hours taught in the Current & Last *Semesters* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Degree | Year | Institution | | | | First | Second |
| 1. | Engr. Dr. OJO Adedayo Olukayode | R33712 | Senior Lecturer | 14/10/2014 | Ph.D. | 2024 | ABUAD | Communication and Intelligent Systems | 11 | Dedicated | 12 | 6 |
| | | | | | M.Eng | 2014 | UPM | | | | | |
| | | | | | B.Tech | 2008 | LAUTECH | | | | | |
| 2. | Engr. Prof. Moses Oluwafemi ONIBONOJE | R30783 | Professor | 19/10/2015 | Ph.D. | 2016 | OAU | Instrumentation and Control, Distributed Systems, Wireless Sensor Network | 12 | Dedicated | 12 | 6 |
| | | | | | M.Eng | 2012 | OAU | | | | | |
| | | | | | B.Eng | 2007 | OAU | | | | | |
| | | | | | HND/ND | 2002 | Federal Polytechnic Ado Ekiti | | | | | |
| 3. | Engr. Prof. Joseph | R5561 | Professor | 1/10/2016 | Ph.D. | 2001 | Uni. Of Duisburg Essen | | 31 | Dedicated | | |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Olufemi DADA | | | | MEng | 2014 | Unilorin | Power and Energy Systems | | | 6 | 4 |
| | | | | | BEng | 1987 | Unilorin | | | | | |
| 4. | Engr. Dr. Ayodeji Olalekan SALAU | R35396 | Associate Professor | 15/02/2013 | Ph.D. | 2018 | OAU | Communication, Electronics and Computer Vision | 12 | Dedicated | 11 | 6 |
| | | | | | M.Eng | 2015 | OAU | | | | | |
| | | | | | B.Eng | 2010 | FUT(minna) | | | | | |
| 5. | Engr. Dr. Joseph Folorunso ORIMOLADE | R12,674 | Senior Lecturer | 19/04/2024 | Ph.D.<br>M.Sc.<br>B.Eng. | (2017)<br>(2004)<br>(1995) | Capetown<br>ABU<br>ABU | Cybersecurity and Communication Systems | 15 | | | |
| 6. | Engr. Dr. ADEJUMOBI Babatunde Segun | R56,341 | Senior Lecturer | 19/10/2020 | Ph.D.<br>M. Sc.<br>B. Sc | 2021<br>2016<br>2012 | South Africa<br>LASU<br>Ibadan | Communication Systems | | | | |
| 7. | Engr. Dr. Oluwafunso Oluwole OSALONI | R54620 | Senior Lecturer | 21/10/2023 | Ph.D. | 2021 | Uni. Kwazulu-natal | Power and Energy Systems | 9 | Dedicated | 8 | 6 |
| | | | | | M.Sc. | 2016 | Uni. Of Cape town | | | | | |
| | | | | | B.Sc | 2012 | FUTA | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8. | Engr. Dr. OGUNLADE Michael Adegoke | R35098 | Senior Lecturer | 19/10/2015 | Ph.D. | 2024 | ABUAD | Communication Electronics and Computer Vision | 10 | Dedicated | 12 | 4 |
| | | | | | M.Eng | 2015 | Tiajun University of technology | | | | | |
| | | | | | B.Sc | 2008 | Uni. Of Ado-Ekiti | | | | | |
| 9. | Engr. Dr. OMOGBAI Nelson Oyakhilomen | R59849 | Lecturer I | 19/04/2024 | Ph.d | 2023 | Nnamdi Azikwe University | Power and Machine | 2 | Dedicated | 11 | 6 |
| | | | | | Sc | 2018 | UNIBEN | | | | | |
| | | | | | B.sc | 2008 | UNIBEN | | | | | |
| 10. | Engr. Dr. Oladapo IBITOYE | R57,893 | Lecturer I | 11/2016 | h.D | | | Artificial Intelligence and Embedded Systems | | | | |
| | | | | | M.Eng. | 2024 | ABUAD | | | | | |
| | | | | | | 2021 | ABUAD | | | | | |
| 11. | Engr. FABUNMI Temitayo Emmanuel | R60980 | Lecturer I | 19/12/2019 | M.Sc. | 2021 | ABUAD | Instrumentation and Control; Electronic Systems | 6 | Dedicated | 10 | 4 |
| | | | | | B.Sc | 2016 | ABUAD | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12. | Engr. AKINOLA Akintunde | R78947 | Lecturer II | 11/05/2023 | M.Sc.<br><br>B.Sc | 2016<br><br>2010 | FUTA<br><br>University of Ado-Ekiti | Communications and Electronic Systems | 8 | Dedicated | 11 | 4 | |
| 13. | Mr. ADEBAJO-AKINPELU Victor | NYR | Assistant Lecturer | 19/04/2024 | M.eng<br><br>B.eng | 2025<br><br>2019 | ABUAD | Control and Instrumentation | 2 | Dedicated | 8 | 6 | |
| 14. | Engr. AWODOSU A. O. | Registered | Lecturer I | | | | | | | Dedicated | | | |

**Technical Staff**

| S/N | Names of Technical Staff | Qualifications | Rank | Employment Status |
|---|---|---|---|---|
| 1. | Mr. ITAMAKINDE Akinyele Abiodun | M.Eng (2022) PGD (2016), HND (2000), OND (1997), | Asst. Chief Technologist | Full time |
| 2. | Mr. OMITOKUN Elijah Olusegun | HND (2009), ND (2006) | Principal Technologist | Full time |
| 3. | Mr. NWAZUOKE Chidiebere Emmanuel | HND (2010), ND (2007) | Principal Technologist | Full time |
| 4. | Mr. KUMUYI Adebayo Gabriel | HND (2011), ND (2008) | Principal Technologist | Full time |
| 5. | Mr. OLANIPEKUN Olayemi Emmanuel | HND (2011), ND (2006) | Technologist II | Full time |

**Administrative Staff**

| S/N | Name of Staff | Rank/Designation | Qualification | Remarks |
|---|---|---|---|---|
| 1 | OLALEYE, Seun Mayowa | Admin Officer II | B.Sc. Ed. Political Science (2016) | Administrative duties |

## COURSE CURRICULUM

### Categorization of Courses

Courses are categorized as follows:

**i. Compulsory Courses:** Students must register for and pass before graduation.

**ii. Elective Courses:** Courses students may offer and pass to obtain the minimum units required for graduation. Such courses are normally selected from within or outside the programme with the approval of the department.

**iii. Required Course:** A course specified by a programme which a student shall take and pass.

**iv. Pre-requisite Courses:** Courses students must take and pass before taking another specified course.

**v. Institutional Courses:** Courses whose knowledge is essential for the preparation of Afe Babalola University for future employment after graduation.

**vi. Carry-over Courses:** On the account of any failure by a student to satisfy a pass requirement in any course, such student will be required to repeat only the course/courses the following academic session or re-write the prescribed examination in such course at the end of the academic session.

Students carrying over courses shall not be allowed to register for more than the approved maximum number of credit for the semester.

### 100 LEVEL FIRST SEMESTER

| Course Code | Course Title | Units | Status | LH | PH |
|---|---|---|---|---|---|
| GST 111 | Communication in English | 2 | C | 15 | 45 |
| GET 101 | Engineer in Society | 1 | C | 15 | - |
| MTH 101 | Elementary Mathematics I: Algebra and Trigonometry | 2 | C | 30 | - |
| PHY 101 | General Physics I: Mechanics | 2 | C | 30 | - |
| PHY 107 | General Practical Physics I | 1 | C | - | 45 |
| CPE 112 | Introduction to Computer Engineering | 2 | C | 30 | - |
| ABUAD-CHM 101 | General Chemistry I | 2 | C | 45 | - |
| ABUAD-CHM 107 | General Practical Chemistry I | 1 | C | - | 45 |
| AFE 101 | Health Awareness Education | 1 | C | 15 | - |
| AFE 103 | Leadership | 1 | C | 15 | - |
| **TOTAL** | | **15** | | | |

**100 LEVEL SECOND SEMESTER**

| Course Code | Course Title | Units | Status | LH | PH |
|---|---|---|---|---|---|
| GST 112 | Nigerian Peoples and Culture | 2 | C | 30 | - |
| GET 102 | Engineering Graphics and Solid Modelling I | 2 | C | 15 | 45 |
| MTH 102 | Elementary Mathematics II: Calculus | 2 | C | 30 | - |
| PHY 102 | General Physics II: Behaviour of Matter | 2 | C | 30 | - |
| PHY 108 | General Practical Physics II | 1 | C | - | 45 |
| ABUAD-CHM 102 | General Chemistry II | 2 | C | 45 | - |
| ABUAD-CHM 108 | General Practical Chemistry II | 1 | C | - | 45 |
| FOREIGN LANGUAGE ELECTIVE (SELECTION OF ONE) | | | | | |
| AFE 124 | French Language I | 1 | C | 15 | - |
| AFE 126 | Chinese Language | 1 | C | 15 | - |
| NIGERIAN LANGUAGE ELECTIVE (SELECTION OF ONE) | | | | | |
| AFE 128 | Igbo Language | 1 | R | 15 | - |
| AFE 130 | Yoruba Language | 1 | R | 15 | - |
| AFE 132 | Hausa Language | 1 | R | 15 | - |
| **TOTAL** | | **14** | | | |

**SECOND YEAR (200 LEVEL) : FIRST SEMESTER COURSE STRUCTURE**

| Course Code | Course Title | Unit | Status | Contact Hours/Week | Total Week Load |
|---|---|---|---|---|---|
| | | | | **L-T-P** | |
| ENG 201 | Engineering Materials | 2 | C | 1-1-0 | 2 |
| ENG 203 | Strength of Materials | 3 | C | 1-1-3 | 5 |
| ENG 221 | Basic Electrical Engineering I | 3 | C | 1-1-3 | 5 |
| ENG 223 | Computer Applications | 2 | C | 1-0-3 | 4 |
| ENG 231 | Engineering Drawing I | 2 | C | 1-1-0 | 2 |
| ENG 233 | Engineering Mechanics I | 2 | C | 1-1-0 | 2 |
| ENG 235 | Engineering Thermodynamics | 2 | C | 1-1-0 | 2 |
| ENG 261 | Workshop Technology I | 2 | C | 0-0-6 | 6 |
| ENG 281 | Engineering Mathematics I | 3 | C | 2-1-0 | 3 |
| GST 201 | Peace and Conflict Studies | 2 | C | 2-0-0 | 2 |
| **One foreign language is required** | | | | | |
| AFE 203* | French Language Acquisition II | 1 | E | 1-0-0 | 1 |
| AFE 205* | Chinese Language II | 1 | E | 1-0-0 | 1 |
| **One Nigerian Language is required** | | | | | |
| AFE 207** | Igbo Language II | 1 | E | 1-0-0 | 1 |
| AFE 209** | Yoruba Language II | 1 | E | 1-0-0 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| AFE 211** | Hausa Language II | 1 | E | 1-0-0 | 1 |
| Total Credits | | 25 | | | |

**SECOND YEAR (200 LEVEL): SECOND SEMESTER COURSE STRUCTURE**

| Course Code | Course Title | Unit | Status | Contact Hours/Week | Total Week Load |
|---|---|---|---|---|---|
| ENG 204 | Introduction to Mechatronics Engineering | 2 | C | 1-1-0 | 2 |
| ENG 214 | Fluid Mechanics | 3 | C | 2-1-0 | 3 |
| ENG 222 | Basic Electrical Engineering II | 3 | C | 1-1-3 | 5 |
| ENG 224 | Computer Programming and Applications | 2 | C | 1-0-3 | 4 |
| ENG 232 | Engineering Drawing II | 2 | C | 1-0-3 | 4 |
| ENG 234 | Engineering Mechanics II | 2 | C | 1-1-0 | 2 |
| ENG 262 | Workshop Technology II | 2 | C | 1-0-3 | 4 |
| ENG 282 | Engineering Mathematics II | 3 | C | 2-1-0 | 3 |
| ENG 284 | Engineer in Society | 2 | C | 2-0-0 | 2 |
| GST 202 | Philosophy and Sustainable Development | 2 | C | 1-1-0 | 2 |
| GST 212 | Introduction to Entrepreneurship | 2 | C | 2-0-0 | 2 |
| AFE 202 | Food Production and Health Awareness | 1 | R | 1-0-0 | 1 |
| **Total Credits** | | **26** | | | |

**300 LEVEL FIRST SEMESTER**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| EEE 313 | Electronics I (Physical) | 3 | 2 | 1 | - | - | C | 3 |
| EEE 315 | Electromagnetic Field and Wave Theory I | 3 | 2 | 1 | - | - | C | 3 |
| EEE 317 | Electrical/Electronic Laboratory I | 1 | - | - | 3 | - | C | 3 |
| EEE 319 | Measurements and Instrumentation | 3 | 2 | 1 | - | - | C | 3 |
| EEE 321 | Electric Circuit Theory I | 3 | 2 | 1 | - | - | C | 3 |
| EEE 325 | Electrical Machines I | 3 | 2 | 1 | - | - | C | 3 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| EEE 341 | Software Development Techniques | 3 | 2 | - | 3 | - | C | 5 |
| ENG 381 | Engineering Mathematics III | 3 | 2 | 1 | - | - | C | 3 |
| AFE 301 | Practical Entrepreneurship | 1 | 1 | - | 3 | - | C | 4 |
| | | **23** | | | | | | |

**300 LEVEL SECOND SEMESTER**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| EEE 314 | Electronics II (Analogue) | 3 | 2 | 1 | - | EEE 313 | C | 3 |
| EEE 316 | Electromagnetic Field and Wave Theory II | 3 | 2 | 1 | - | EEE 315 | C | 3 |
| EEE 318 | Electrical/Electronic Laboratory II | 1 | - | - | 3 | - | C | 3 |
| EEE 322 | Electric Circuit Theory II | 3 | 2 | 1 | - | EEE 321 | C | 3 |
| EEE 324 | Linear Systems | 3 | 2 | 1 | - | - | C | 3 |
| EEE 326 | Electrical Machines II | 3 | 2 | 1 | - | EEE 325 | C | 3 |
| EEE 328 | Communication Principles | 3 | 2 | 1 | - | | C | 3 |
| ENG 382 | Engineering Mathematics IV | 3 | 2 | 1 | - | | C | 3 |
| ENG 384 | Engineering Economics | 2 | 2 | - | - | | C | 2 |
| | | **24** | | | | | | |

**400 LEVEL FIRST SEMESTER**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| EEE 405 | Numerical Computations | 3 | 2 | 1 | - | - | C | 3 |
| EEE 415 | Assembly Language Programming | 2 | 2 | - | - | - | C | 2 |
| EEE 423 | Electrical/Electronic Laboratory III | 1 | - | - | 3 | EEE 318 | C | 3 |
| EEE 427 | Data Communication and Networks | 3 | 2 | 1 | - | - | C | 3 |
| EEE 431 | Electric Power Principles | 3 | 2 | 1 | - | - | C | 3 |
| EEE 441 | Control Systems and Servomechanism | 3 | 2 | 1 | - | - | C | 3 |
| EEE 471 | Digital Electronics and Systems | 3 | 2 | 1 | - | EEE 314 | C | 3 |
| ENG 485 | Technical Communications (Report Writing) | 2 | 2 | - | - | -- | C | 2 |
| | | **20** | | | | | | |

**400 LEVEL SECOND SEMESTER**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| ENG 290 | Student Work Experience Programme I (SWEP I) | 3 | | | | - | C | |
| ENG 390 | Student Work Experience Programme II (SWEP II) | 3 | | | | ENG 290 | C | |
| ENG 490 | Student Industrial Work Experience Scheme (SIWES) | 6 | | | | ENG 390 | C | |
| | | **12** | | | | | | |

**500 LEVEL FIRST SEMESTER**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| EEE 511 | Digital Signal Processing | 2 | 2 | - | - | - | C | 2 |
| EEE 521 | Computer Aided Design & Simulation | 2 | 2 | - | - | - | C | 2 |
| EEE 531 | Solid State Electronics | 3 | 2 | 1 | - | - | C | 3 |
| EEE 541 | Reliability of Electrical & Electronic System | 3 | 2 | 1 | - | | C | 3 |
| EEE 551 | Power Electronics & Drives | 3 | 2 | 1 | - | EEE 326 | C | 3 |
| EEE 571 | Student Project | 3 | - | - | 9 | - | C | 9 |
| **Students to offer minimum of Two (2) courses** | | | | | | | | |
| EEE 535 | Microwave & Satellite Communications | 3 | 3 | - | - | - | E | 3 |
| EEE 537 | Antenna Theory and Radar Systems | 3 | 2 | 1 | - | - | E | 3 |
| EEE 553 | Power System Engineering I | 3 | 3 | - | - | **-** | E | 3 |
| EEE 555 | High Voltage Engineering | 3 | 2 | 1 | - | **-** | E | 3 |
| EEE 564 | Modern Control System | 3 | 3 | - | - | - | E | 3 |
| EEE 561 | Process Control and Automation | 3 | 2 | 1 | - | - | E | 3 |
| | | **22** | | | | | | |

**SECOND SEMESTER 500 LEVEL COURSES**

| Course Code | Course Title | Units | Duration (Hours) | | | Pre-Req. | Status | Total Week Load |
|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | | |
| EEE 512 | Digital Communication | 3 | 3 | - | - | - | C | 3 |
| EEE 522 | Microprocessor Application, Organization & Embedded Systems | 3 | 3 | - | - | - | C | 3 |
| EEE 524 | Analog IC Designs & Applications | 2 | 2 | - | - | - | C | 2 |
| EEE 552 | Industrial Utilization of Electric Power | 2 | 2 | - | - | - | C | 2 |
| EEE 564 | Current trends in Electrical/Electronic Engineering | 1 | 1 | - | - | - | C | 1 |
| EEE 572 | Student Project | 3 | - | - | 9 | EEE 571 | C | 9 |
| **Students to offer minimum of Two (2) courses** | | | | | | | | |
| EEE 536 | Wireless Communication | 3 | 3 | - | - | | E | 3 |
| EEE 576 | Microcomputer Technology | 3 | 3 | - | - | - | E | 3 |
| EEE 554 | Power System Engineering II | 3 | 3 | - | - | EEE 553 | E | 3 |
| EEE 556 | Machine Application and Protection | 3 | 3 | - | - | | E | 3 |
| EEE 566 | Introduction to Non-Linear Systems | 3 | 3 | - | - | - | E | 3 |
| EEE 578 | Introduction to Biomedical | 3 | 3 | - | - | - | E | 3 |
| | | **20** | | | | | | |

## COURSE DESCRIPTION

### 100 LEVEL FIRST SEMESTER

### GST 111: Communication in English (2 Units C: LH=15; PH=45)

**Learning Outcomes**

At the end of this course, students should be able to:

1. identify possible sound patterns in English Language;
2. list notable language skills;
3. classify word formation processes;
4. construct simple and fairly complex sentences in English;
5. apply logical and critical reasoning skills for meaningful presentations;
6. demonstrate an appreciable level of the art of public speaking and listening; and
7. write simple and technical reports.

**Course Contents**

Sounds and sound patterns in English Language (vowels and consonants, phonetics and phonology). English word classes (lexical and grammatical words, definitions, forms, functions, usages, collocations). major word formation processes; the sentence in English (types: structural and functional). grammar and usage (tense, concord and modality). Reading and types of reading, comprehension skills, 3RsQ. Logical and critical thinking; reasoning methods (logic and syllogism, inductive and deductive argument, analogy, generalisation and explanations). Ethical considerations, copyright rules and infringements. Writing activities: pre-writing (brainstorming and outlining). writing (paragraphing, punctuation and expression). post- writing (editing and proofreading). Types of writing (summary, essays, letter, curriculum vitae, report writing, note-making) etc. Mechanics of writing. Information and Communication Technology in modern language learning. Language skills for effective communication. The art of public speaking.

### GET 101: Engineer in Society (1 Unit C: LH=15)

**Learning Outcomes**

At the end of this course, the students should be able to:

1. differentiate between science, engineering and technology, and relate them to innovation;
2. distinguish between the different cadres of engineering – engineers, technologists, technicians and craftsmen and their respective roles and competencies;
3. identify and distinguish between the relevant professional bodies in engineering;
4. categorise the goals of global development or sustainable development goals (SDGs); and
5. identify and evaluate safety and risk in engineering practice.

**Course Contents**

History, evolution and philosophy of science. engineering and technology. The engineering profession – engineering family (engineers, technologists, technicians and craftsmen), professional bodies and societies. Engineers' code of conduct and ethics, and engineering literacy. Sustainable development goals (SDGs), innovation, infrastructures and nation building - economy, politics, business. Safety and risk analysis in engineering practice. Engineering competency skills – curriculum overview, technical, soft and digital skills. Guest seminars and invited lectures from different engineering professional associations.

**MTH 101: Elementary Mathematics I (Algebra and Trigonometry)**
**(2 Units C: LH=30)**

**Learning Outcomes**
At the end of the course students should be able to:
1. define and explain set, subset, union, intersection, complements, and demonstrate the use of Venn diagrams;
2. solve quadratic equations;
3. solve trigonometric functions;
4. identify various types of numbers; and
5. solve some problems using binomial theorem.

**Course Contents**
Elementary set theory, subsets, union, intersection, complements, Venn diagrams. Real numbers, integers, rational and irrational numbers. Mathematical induction, real sequences and series, theory of quadratic equations, binomial theorem, complex numbers, algebra of complex numbers, the argand diagram. De-Moiré's theorem, nth roots of unity. Circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

**PHY 101: General Physics I (Mechanics) (2 Units C: LH=30)**

**Learning Outcomes**
On completion, the students should be able to:
1. identify and deduce the physical quantities and their units;
2. differentiate between vectors and scalars;
3. describe and evaluate motion of systems on the basis of the fundamental laws of mechanics;
4. apply Newton's laws to describe and solve simple problems of motion;
5. evaluate work, energy, velocity, momentum, acceleration, and torque of moving or rotating objects;
6. explain and apply the principles of conservation of energy, linear and angular momentum;
7. describe the laws governing motion under gravity; and
8. explain motion under gravity and quantitatively determine behaviour of objects moving under gravity.

**Course Contents**
Space and time; units and dimension, vectors and scalars, differentiation of vectors: displacement, velocity and acceleration; kinematics; Newton's laws of motion (inertial frames, impulse, force and action at a distance, momentum conservation); relative motion; application of Newtonian mechanics; equations of motion; conservation principles in physics, conservative forces, conservation of linear momentum, kinetic energy and work, potential energy, system of particles, centre of mass; rotational motion; torque, vector product, moment, rotation of coordinate axes and angular momentum. Polar coordinates; conservation of angular momentum; circular motion; moments of inertia, gyroscopes and precession; gravitation: Newton's law of gravitation, Kepler's laws of planetary motion, gravitational potential energy, escape velocity, satellites motion and orbits.

**PHY 107: General Practical Physics I (1 Unit C: PH 45)**

**Learning Outcomes**
On completion, the student should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs; and
5. draw conclusions from numerical and graphical analysis of data.

**Course Contents**
This introductory course emphasizes quantitative measurements. Experimental techniques. The treatment of measurement errors. Graphical analysis. The experiments include studies of meters, the oscilloscope, mechanical systems, electrical and mechanical resonant systems, light, heat, viscosity, etc. (covered in PHY 101, 102, 103 and PHY 104). However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis, and deduction.

**CPE 112: Introduction to Computer Engineering (2 Units C: LH=30)**

**Learning Outcomes**
At the end of the course the student should be able to:
1. represent and manipulate information in binary form in a written and readable form;
2. design, physically implement, and debug basic combinational and sequential logic circuits;
3. write structural and data flow models of logic circuits in a hardware description language;
4. implement designs represented in a register transfer language;
5. discuss and explain the organisation and operation of a basic digital computer;
6. describe the execution of machine language computer programmes by a basic computer in writing or flow chart;

7. write elementary assembly language programmes and discuss their translation to machine language programmes;
8. use Boolean algebra or K-maps to simplify complex Boolean expressions;
9. convert numbers between any two number systems, especially decimal, binary, octal and hex and represent sign numbers;
10. design arithmetic circuits to perform addition and subtraction of signed numbers and detect overflow conditions;
11. implement functions using AND/OR gates, OR/AND gates, NORs only, NANDs only, multiplexers or decoders.
12. design the basic flip flops using sequential logic;
13. design, implement and test a simple circuit based on a specified word problem;
14. programme simple microcontrollers in assembly language; and
15. identify a local environment-related need, write and present group reports on hardware and software design projects using cooperative learning approach (learning team work in problem solving and improvisation).

**Course Contents**
Historical development of modern computing and computer engineering profession; roles and responsibilities of the computer engineer; career paths and development (public and private sectors, academic/research and industry); overview of computer engineering design; computer devices/hardware in the age of' smartness' and Internet of Things and People 'IoTs and P'; identification of computer software and hardware components and operational relationships (central processing units, input/output devices, operating systems, languages,

**ABUAD-CHM 101: General Chemistry I (2 Units C: LH=30)**

**Senate-approved relevance**
This course is essential for students at our university as it builds a strong foundation in chemistry, aligning with Sustainable Development Goals (SDGs) 4 (Quality Education) and 9 (Industry, Innovation, and Infrastructure). A deep understanding of chemical principles is crucial for advancing in scientific research, environmental sustainability, and technological innovation. By mastering the content, students will be equipped to contribute to sustainable industrial practices, environmental conservation, and scientific advancements, directly impacting global challenges. This course fosters critical thinking and problem-solving skills, preparing students to be leaders in science, technology, and sustainability.

**Overview**
This course introduces the fundamental concepts of chemistry, including atoms, molecules, elements, and compounds. It explores modern electronic theory, electronic configuration, and periodicity, and examines molecular structures, chemical bonding, and stoichiometry. Students will also study the kinetic theory of matter, thermochemistry, reaction rates, equilibrium, and thermodynamics. Additionally, the course covers the properties of gases, acids, bases, redox

reactions, electrochemistry, and radioactivity, providing a comprehensive understanding of essential chemical principles.

**Objectives**

1. understand and describe the basic structure of atoms, molecules, elements, and compounds.
2. apply electronic configuration and periodicity to explain chemical properties.
3. analyze and predict the shapes and structures of molecules and solids.
4. solve chemical equations and apply principles of stoichiometry, thermodynamics, and kinetics.
5. understand and apply concepts of redox reactions, electrochemistry, and radioactivity.

**Learning Outcomes**

At the end of this course, the students should be able to:
1. define atom, molecules and chemical reactions;
2. discuss the modern electronic theory of atoms;
3. write electronic configurations of elements on the periodic table;
4. rationalise the trends of atomic radii, ionisation energies, electronegativity of the elements, based on their position in the periodic table;
5. identify and balance oxidation–reduction equation and solve redox titration problems;
6. draw shapes of simple molecules and hybridised orbitals;
7. identify the characteristics of acids, bases and salts, and solve problems based on their quantitative relationship;
8. apply the principles of equilibrium to aqueous systems using LeChatelier's principle to predict the effect of concentration, pressure and temperature changes on equilibrium mixtures;
9. analyse and perform calculations with the thermodynamic functions, enthalpy, entropy and free energy; and
10. determine rates of reactions and its dependence on concentration, time and temperature.

**Course Contents**

Atoms, molecules, elements and compounds, and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridisation and shapes of simple molecules. Valence forces; Structure of solids. Chemical equations and stoichiometry; chemical bonding and intermolecular forces, kinetic theory of matter. Elementary thermochemistry; rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox reactions and introduction to electrochemistry. Radioactivity.

**ABUAD-CHM 107: General Practical Chemistry I (1 Unit C: PH 45)**

**Senate-approved relevance**

This laboratory course is integral to our university's commitment to Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It provides hands-on experience in fundamental chemical techniques, reinforcing theoretical knowledge from CHM 101. By mastering

these practical skills, students develop the ability to conduct accurate experiments, analyze data, and present findings effectively—essential competencies for careers in science, industry, and research. The course also promotes safe and sustainable laboratory practices, preparing students to contribute to sustainable industrial processes and environmental stewardship.

**Overview**

This laboratory course offers practical experience in key chemical techniques, complementing the theoretical content of CHM 101. Students will engage in acid-base titrations, qualitative analysis, redox reactions, and gravimetric analysis. Emphasis is placed on accurate data collection, analysis, and presentation, fostering essential laboratory skills. The course prepares students to apply these techniques in scientific research and industry, ensuring a comprehensive understanding of experimental chemistry.

**Objectives**

The objectives of the course are to:

1. perform acid-base titrations with precision and accuracy.
2. conduct qualitative and quantitative analysis of chemical substances.
3. understand and execute redox reactions and gravimetric analysis.
4. analyze and interpret experimental data effectively.
5. present laboratory findings clearly and professionally.

**Learning Outcomes**

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correct carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. state the differences between primary and secondary standards;
5. perform redox titration;
6. record observations and measurements in the laboratory notebooks; and
7. analyse the data to arrive at scientific conclusions.

**Course Contents**

Laboratory experiments designed to reflect topics presented in courses CHM 101 and CHM 102. These include acid-base titrations, qualitative analysis, redox reactions, gravimetric analysis, data analysis and presentation.

**100 LEVEL SECOND SEMESTER**

**GST 112: Nigerian Peoples and Cultures (2 Units C: LH=30)**

**Learning Outcomes**

At the completion of the course, the students are expected to be able:

1. analyse the historical foundation of Nigerian cultures and arts in pre-colonial times;
2. identify and list the major linguistic groups in Nigeria;

3. explain the gradual evolution of Nigeria as a political entity;
4. analyse the concepts of trade and economic self-reliance of Nigerian peoples in relation to national development;
5. enumerate the challenges of the Nigerian state regarding nation building;
6. analyse the role of the Judiciary in upholding fundamental human rights
7. identify the acceptable norms and values of the major ethnic groups in Nigeria; and
8. list possible solutions to identifiable Nigerian environmental, moral and value problems.

**Course Contents**

Nigerian history, culture and art up to 1800 (Yoruba, Hausa and Igbo peoples and cultures; peoples and cultures of the minority ethnic groups). Nigeria under colonial rule (advent of colonial rule in Nigeria; colonial administration of Nigeria). Evolution of Nigeria as a political unit (amalgamation of Nigeria in 1914; formation of political parties in Nigeria; nationalist movement and struggle for independence). Nigeria and challenges of nation building (military intervention in Nigerian politics; Nigerian Civil War). Concepts of trade and economics of self-reliance (indigenous trade and market system; indigenous apprenticeship system among Nigerian peoples; trade, skill acquisition and self-reliance). Social justice and national development (definition and classification of law); Judiciary and fundamental rights. Individuals, norms and values (basic Nigerian norms and values, patterns of citizenship acquisition; citizenship and civic responsibilities; indigenous languages, usage and development; negative attitudes and conducts [Cultism, kidnapping and other related social vices]). Re-orientation, moral and national values (The 3Rs – Reconstruction, Rehabilitation and Re-orientation; re-orientation strategies: Operation Feed the Nation (OFN), Green Revolution, Austerity Measures, War Against Indiscipline and Corruption (WAIC), Mass Mobilization for Self-Reliance, Social Justice and Economic Recovery (MAMSER), National Orientation Agency (NOA). Current socio-political and cultural developments in Nigeria.

**GET 102: Engineering Graphics and Solid Modelling I (2 Units C: LH=15; PH=45)**

**Learning Outcomes**

At the end of this course, the students should be able to:

1. have a good grasp of design thinking and be obsessed with the determination to apply such to solving simple everyday and also complex problems;
2. recognise the fundamental concepts of engineering drawing and graphics;
3. show skills to represent the world of engineering objects in actionable solid models, and put such models in a form where they can be inputs for simulation and analyses;
4. analyse such models for strength and cost.
5. prepare the objects for modern production and manufacturing techniques of additive and subtractive manufacturing;

6. recognise that engineering is multidisciplinary in the sense that mechanical, electrical and other parts of physical structures are modelled in context as opposed to the analytical nature of the courses they take;and
7. analyse and master the basics of mechanical and thermal loads in engineering systems.

**Course Contents**

Introduction to design thinking and engineering graphics. First and third angle orthogonal projections. Isometric projections; sectioning, conventional practices, conic sections and development. Freehand and guided sketching – pictorial and orthographic. Visualisation and solid modelling in design, prototyping and product-making. User interfaces in concrete terms. Design, drawing, animation, rendering and simulation work spaces. Sketching of 3D objects. Viewports and sectioning to shop drawings in orthographic projections and perspectives. Automated viewports. Sheet metal and surface modelling. Material selection and rendering. This course will use latest professional design tools such as fusion 360, solid works, solid edge or equivalent.

**MTH 102: Elementary Mathematics II (Calculus) (2 Units C: LH=30)**

**Learning Outcomes**

At the end of the course, students should be able to:
1. identify the types of rules in differentiation and integration;
2. recognise and understand the meaning of function of a real variable, graphs, limits and continuity;
3. solve some applications of definite integrals in areas and volumes;
4. solve function of a real variable, plot relevant graphs, identify limits and idea of continuity;
5. identify the derivative as limit of rate of change;
6. identify techniques of differentiation and perform extreme curve sketching;
7. identify integration as an inverse of differentiation;
8. identify methods of integration and definite integrals; and
9. perform integration application to areas, volumes.

**Course Contents**

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration, definite integrals, reduction formulae, application to areas, volumes (including approximate integration: Trapezium and Simpson's rule).

**PHY 102: General Physics II (Behaviour of Matter) (2 Units C: LH=30)**

**Learning Outcomes**

On completion, the students should be able to:
1. explain the concepts of heat and temperature and relate the temperature scales;
2. define, derive and apply the fundamental thermodynamic relations to thermal systems;

3. describe and explain the first and second laws of thermodynamics, and the concept of entropy;
4. state the assumptions of the kinetic theory and apply techniques of describing macroscopic behaviour;
5. deduce the formalism of thermodynamics and apply it to simple systems in thermal equilibrium; and
6. describe and determine the effect of forces and deformation of materials and surfaces.

**Course Contents**
Heat and temperature, temperature scales; gas laws; general gas equation; thermal conductivity; first Law of thermodynamics; heat, work and internal energy, reversibility; thermodynamic processes; adiabatic, isothermal, isobaric; second law of thermodynamics; heat engines and entropy, Zero's law of thermodynamics; kinetic theory of gases; molecular collisions and mean free path; elasticity; Hooke's law, Young's shear and bulk moduli; hydrostatics; pressure, buoyancy, Archimedes' principles; Bernoullis equation and incompressible fluid flow; surface tension; adhesion, cohesion, viscosity, capillarity, drops and bubbles.

**PHY 108: General Practical Physics II (1 Unit C: PH 45)**

**Learning Outcomes**
On completion, the student should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs;
5. draw conclusions from numerical and graphical analysis of data; and
6. prepare and present practical reports.

**Course Contents**
This practical course is a continuation of PHY 107 and is intended to be taught during the second semester of the 100 level to cover the practical aspect of the theoretical courses that have been covered with emphasis on quantitative measurements, the treatment of measurement errors, and graphical analysis. However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis and deduction.

**ABUAD-CHM 102: General Chemistry II (2 Units C: LH=30)**

**Senate-approved relevance**
This course is crucial to our university's mission of advancing Quality Education (SDG 4) and fostering Innovation (SDG 9). It provides a comprehensive understanding of organic chemistry, which is fundamental to multiple scientific disciplines, including pharmaceuticals, materials science, and environmental science. The course highlights the role of organic chemistry in developing new materials like fullerenes, which are essential for nanotechnology. By mastering the concepts and techniques in organic chemistry, students are prepared to contribute to innovative

solutions in health, industry, and sustainability, addressing global challenges through scientific advancement.

**Overview**

This course offers an in-depth study of organic chemistry, beginning with its historical development and significance. It covers the electronic theory, structure determination, nomenclature, and functional groups of organic compounds. Students will explore reaction mechanisms, kinetics, stereochemistry, and the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds. Additionally, the course introduces the chemistry of selected metals and non-metals, including comparative chemistry of group IA, IIA, and IVA elements, and transition metal chemistry, with a focus on their practical applications in various industries.

**Objectives**

1. understand the historical development and significance of organic chemistry.
2. describe the structure, nomenclature, and functional groups of organic compounds.
3. analyze reaction mechanisms, kinetics, and stereochemistry in organic reactions.
4. explore the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds.
5. examine the chemistry of selected metals, non-metals, and transition metals, including their industrial applications.

**Learning Outcomes**

**At the end of this course, the students should be able to:**

1. state the importance and development of organic chemistry;
2. define fullerenes and its applications;
3. discuss electronic theory;
4. determine the qualitative and quantitative of structures in organic chemistry;
5. state rules guiding nomenclature and functional group classes of organic chemistry;
6. determine the rate of reaction to predict mechanisms of reaction;
7. identify classes of organic functional group with brief description of their chemistry;
8. discuss comparative chemistry of group 1A, IIA and IVA elements; and
9. describe basic properties of transition metals.

**Course Contents**

Historical survey of the development and importance of organic chemistry; fullerenes as fourth allotrope of carbon, uses as nanotubules, nanostructures, nanochemistry. Electronic theory in organic chemistry. Isolation and purification of organic compounds; determination of structures of organic compounds including qualitative and quantitative analysis in organic chemistry; nomenclature and functional group classes of organic compounds. Introductory reaction mechanism and kinetics. Stereochemistry. The chemistry of alkanes, alkenes, alkynes, alcohols, ethers, amines, alkyl halides, nitriles, aldehydes, ketones, carboxylic acids and derivatives. The

chemistry of selected metals and non-metals. Comparative chemistry of group IA, IIA and IVA elements. Introduction to transition metal chemistry.

**ABUAD- CHM 108: General Practical Chemistry II (1 Unit C: PH 45)**

**Senate-approved relevance**

This practical course is integral to our university's focus on Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It builds on the foundation established in CHM 107, enhancing students' proficiency in functional group analysis and quantitative analysis using volumetric methods. These skills are essential for accurate chemical analysis in research and industry, promoting sustainable practices in chemical production and environmental monitoring. By mastering these techniques, students are better prepared to contribute to scientific innovation and responsible resource management, aligning with global sustainability goals.

**Overview**

This course continues from CHM 107, offering advanced laboratory experiments in functional group analysis and quantitative analysis through volumetric methods. Students will refine their skills in identifying functional groups and conducting precise quantitative measurements. The course emphasizes accurate data collection, analysis, and interpretation, which are critical for success in research and industrial applications. By the end of the course, students will have a deeper understanding of practical chemical analysis and its importance in various scientific fields.

**Objectives**

**The objectives of the course are to:**

1. perform advanced functional group analysis accurately.
2. conduct quantitative analysis using volumetric methods with precision.
3. analyze and interpret experimental data effectively.
4. develop proficiency in laboratory techniques for chemical analysis.
5. apply practical skills to real-world chemical problems and research.

**Learning Outcomes**

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correctly carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. identify and carry out preliminary tests which include ignition, boiling point, melting point, test on known and unknown organic compounds;
5. carry out solubility tests on known and unknown organic compounds;
6. carry out elemental tests on known and unknown compounds; and
7. carry out functional group/confirmatory test on known and unknown compounds which could be acidic/basic/ neutral organic compounds.

**Course Contents**
Continuation of CHM 107. Additional laboratory experiments to include functional group analysis, quantitative analysis using volumetric methods.

**200 LEVEL FIRST SEMESTER**

**ENG 201 Engineering Materials (2 Units)**
**Learning Outcomes**

1. Classify various engineering materials (metals, ceramics, composites, and polymers) based on their atomic structure and inter-atomic bonding.
2. Explain the relationship between the microstructure of materials and their physical/mechanical properties.
3. Analyze material behavior in common environments to determine suitability for specific engineering applications.
4. Describe the fabrication processes and alloying theories of common engineering materials.
5. Interpret simple phase diagrams, such as the Iron-Carbon (Fe-C) cooling curve, to predict material solidification and properties.

**Course Contents**

Classes of engineering materials; metals, ceramics, composite, polymer. Introduction to electronic configuration, atomic structures, inter-atomic bonding mechanisms, crystal and microstructure. Relationships between structure and properties of metals, alloys, ceramics and plastics. Principles of the behaviour of materials in common environments. Fabrication processes and applications. Alloying theory and solidification cooling curves illustrated by simple phase diagram e.g. Fe-C diagram.

**ENG 203 Strength of Materials (3 Units)**

**Learning Outcomes**

1.Calculate stresses and strains in bars of varying sections and composite members under different loading conditions.

2.Apply generalized Hooke's law to evaluate structural responses to mechanical loading and temperature changes.

3.Analyze the shear force and bending moments of beams and construct accurate diagrams for symmetrical and combined loadings.

4.Determine stress and strain transformations using analytical equations and Mohr's circle.

5.Evaluate torsion in circular members and determine the elastic buckling characteristics of columns.

**Course Contents**

Simple Stresses and Strains, Stresses and Strains in Bars of Varying Sections. Consideration of equilibrium; composite members, stress-strain relation. Generalized Hooke's law. Stresses and strains due to loading and temperature changes. Elastic constants, centre of gravity, moment of inertia. Torsion of circular members. Shear force, bending moments and bending stresses in beams with symmetrical and combined loadings. Stress and strain transformation equations and Mohr's circle. Elastic buckling of columns.

**ENG 221 Basic Electrical Engineering I (3 Units)**

**Learning Outcomes**

1. Evaluate the interactions between electric fields of charge and magnetic fields of current, assessing their implications in basic engineering applications.
2. Explain the characteristics of lumped circuit elements (resistance, capacitance, inductance) and their terminal behavior.
3. Apply network theorems (Kirchhoff's laws, Superposition, Thevenin's, Norton's, and Star/Delta transformations) to solve practical electrical circuit problems.
4. Analyze single-phase and three-phase AC circuits, including active loads, apparent power, and both balanced and unbalanced conditions.
5. Apply and interpret features of magnetic circuits, including magnetomotive force, permeability, and B–H curves of materials, in solving practical problems.

**Course Contents**

Fundamental concepts - Electric fields, charges, magnetic fields. current, B - H curves Kirchhoff's laws, superposition. Thevenin, Norton theorems, Reciprocity, RL, RC, RLC circuits. DC, AC bridges, Resistance, Capacitance, Inductance measurement, Transducers, Single phase circuits, Complex J - notion, AC circuits, impedance, admittance, susceptance.

**ENG 223 Computer Applications (2 Units)**

**Learning Outcomes**

1. Identify the functions and appropriate use of Personal Computer (PC) components and peripheral devices.

2. Apply word processing, spreadsheet, and presentation software packages to solve basic data management and reporting tasks.

3. Utilize database management packages to organize and retrieve data effectively.

4. Demonstrate safe operation, preventive maintenance, and proper filing systems for personal computers.

5. Execute a mini-project that synthesizes multiple software packages to demonstrate proficiency in computer applications.

**Course Contents**

Identification, function, applications, and use of PC parts and peripheral devices. Safety precautions and preventive maintenance of PC. Filing system. Word processing applications and use. Internet: available services, principle of operation, applications and demonstrations. Spreadsheet: applications and use. Database management packages: applications and demonstrations. Report presentation software packages: applications, demonstrations, and use. Mini-project to test proficiency in use of these software packages.

**ENG 231 Engineering Drawing I (2 Units)**

**Learning Outcomes**

1. Interpret and generate multi-view representations (1st and 3rd angle projections) of engineering components.

2. Construct isometric and oblique drawings (Cavalier and Cabinet) of moderately complex objects.

3. Apply standard dimensioning, sectioning, and conventional practices to mechanical drawings.

4. Create accurate auxiliary views for complex geometries.

5. Accurately represent hardware elements such as threads, bolted joints, keys, and cottered joints using standard conventions.

**Course Contents**

Revision of multi-view representation. Harder examples on two and three view representation (1st and 3rd angles). Harder examples on isometric drawing to include simple pictorial assembly drawing in isometric. Harder examples on oblique drawing (Cavalier, Cabinet and Angles other than 45 degrees). Dimensioning. Sections and Conventions. Auxiliary views. Representation and specification of threads. Bolted joints. Keys and cottered joints. Conventional representations (see BS 308).

**ENG 233 Engineering Mechanics I (2 Units)**

**Learning Outcomes**

1. Resolve forces, moments, and couples acting on structural components and machine parts.

2. Apply the principles of equilibrium to analyze simple structures.

3. Calculate the first and second moments of area and determine the centroids of composite shapes.

4. Analyze the kinematics of particles and rigid bodies moving in plane motion.

5. Apply Newton's laws of motion, kinetic energy, and momentum principles to solve dynamic engineering problems.

**Course Contents**

Forces, moments, couples. Equilibrium of simple structures and machine parts. Friction. First and second moments of area; centroids. Kinematics of particles and rigid bodies in plane motion. Newton's laws of motion. Kinetic energy and momentum analyses.

**ENG 235 Engineering Thermodynamics I (2 Units)**

**Learning Outcomes**

1. Define fundamental thermodynamic concepts including systems, surroundings, temperature, work, and heat.

2. Apply the Zeroth, First, Second, and Third laws of thermodynamics to analyze engineering processes.

3. Evaluate the behavior of pure substances and ideal gases under various thermodynamic cycles.

4. Analyze the performance of heat engines, heat pumps, and refrigeration systems using entropy and combined laws.

5. Apply Maxwell's relations and PVT relationships to analyze two-phase systems and thermodynamic solutions.

**Course Contents**

Basic concepts such as definition of terms and general concepts of systems, surrounding process, temperature, heat, work and energy, quantitative relations of Zeroth, first, second and third laws of thermodynamics. Behaviour of pure substances and perfect gases. Ideal gas cycles. First law of thermodynamics, application to heat engines. Entropy and the second law, combined first and Second laws. Ideal gases Joule-Thomson's coefficient. Equilibrium processes. Maxwell's relation. Two phase systems. Thermodynamic functions of solution. P.V.T. relationship. Work from heat pump refrigeration.

**ENG 261 Workshop Technology I (2 Units)**

**Learning Outcomes**

1. Demonstrate strict adherence to workshop safety rules, identifying hazards and using protective gear appropriately.

2. Select and operate appropriate hand and powered tools for woodwork fitting and joining projects.

3. Perform fundamental machining operations including turning, drilling, grinding, and milling.

4. Execute basic sheet metal fabrication techniques such as rolling, bending, and shearing.

5. Describe the setup and principles of domestic and industrial electrical wiring systems.

**Course Contents**

Introduction to practices and skills in general engineering through instruction in operation of hand and powered tools for wood and metal cutting and fabrication. Supervised hands - on experience in safe usage of tools and machines for selected tasks.

i. Laboratory and Workshop: (Basic hours) - Safety rules and regulations. Sources of accidents and prevention, safety gears, safe working environment/condition, ergonomics, laboratory and workshop layouts, working procedures, firefighting equipment and operations, Lighting and signs, etc. Electrical trades (Power, Electronics, telecommunication), Civil Trades (Building Construction, Roads, Dams, Hydraulics, Sanitation, etc.)
ii. Benchwork: (9 Hours) – Introduction to workshop tools. Use and care of hand tools. Measuring instruments and their uses and equipment, bench fittings. Practical project.
iii. Woodwork: (9 Hours) – Wood work machines and working principles, operation of machines. Simple wood work project for fitting and joining.
iv. Machining: (18 Hours) –Introduction to machine tools, their general characteristics. Simple operations of turning, drilling, grinding, milling, threading, horning, buffing, reaming. Introduction to turret lathes, milling machines (horizontal, vertical and universal), Grinding machines (surface grinder, internal grinding machine, external grinding machine, universal grinding machine, shaping machine) Simple practical project)
v. Fabrication: (9 Hours) –Introduction to fabrication machines, rolling, bending, ribbing, notching, shearing, etc. Operation of machines. Simple project on metal fabrication.
vi. Electrical Wiring: Electrical wiring system: domestic and industrial setup.

**ENG 281 Engineering Mathematics I (3 Units)**

**Learning Outcomes**

1. Evaluate limits, continuity, and derivatives (partial and total) of complex functions.

2. Formulate and solve linear first-order differential equations.

3. Perform mathematical operations using matrices, determinants, and composite functions.

4. Apply vector algebra and calculus (including directional derivatives) to physical engineering variables.

5. Utilize mathematical software tools (Microsoft Excel, MATLAB, and/or MathCAD) to solve engineering math problems.

**Course Contents**

Limits, Continuity, differentiation, introduction to linear first order differential equations, partial and total derivatives, composite functions, matrices and determinants, Vector algebra, Vector calculus, Directional Derivatives. Applications of Microsoft Excel, MATLAB and/or MathCAD.

**GST 201 Peace and Conflict Studies (2 Units)**

Basic Concepts in peace studies and conflict resolution; Peace as vehicle of unity and development; Conflict issues; Types of conflict, e. g. Ethnic/religious/political/ economic conflicts; Root causes of conflicts and violence in Africa; Indigene/settler phenomenon; Peace – building; Management of conflict and security. Elements of peace studies and conflict resolution; Developing a culture of peace; Peace mediation and peace-keeping; Alternative Dispute Resolution (ADR). Dialogue/arbitration in conflict resolution; Role of international organizations in conflict resolution, e.g. ECOWAS, African Union, United Nations, etc.

**AFE 203 French Language Acquisition II (1 Unit)**
Introduction to French, Alphabets and numeracy for effective communication (written and oral), Conjugation and simple sentence construction based on communication approach, Sentence construction, Comprehension and reading of simple texts.

**AFE 205 Chinese Language II (1 Unit)**
Introduction to Chinese, Alphabets and numeracy for effective communication (written and oral), Conjugation and simple sentence construction based on communication approach, Sentence construction, Comprehension and reading of simple texts.

**AFE 207 Igbo Language II (1 Unit)**

Introduction to Igbo, Alphabets and numeracy for effective communication (written and oral), Conjugation and simple sentence construction based on communication approach, Sentence construction, Comprehension and reading of simple texts.

**AFE 209 Yoruba Language II (1 Unit)**
Introduction to Yoruba, Alphabets and numeracy for effective communication (written and oral), Conjugation and simple sentence construction based on communication approach, Sentence construction, Comprehension and reading of simple texts.

**AFE 211 Hausa Language II (1 Unit)**
Introduction to Hausa, Alphabets and numeracy for effective communication (written and oral), Conjugation and simple sentence construction based on communication approach, Sentence construction, Comprehension and reading of simple texts.

**200 LEVEL SECOND SEMESTER**

**ENG 204 Introduction to Mechatronics Engineering (2 Units)**

**Learning Outcomes**

1. Analyze basic DC and AC circuits containing fundamental electrical components and semiconductors.

2. Select appropriate sensors and transducers for measuring displacement, velocity, force, pressure, temperature, and light.

3. Design basic signal conditioning operations including amplification, filtering, and protection.

4. Evaluate pneumatic, hydraulic, and mechanical actuation systems for specific mechatronic applications.

5. Explain the operational principles of electrical actuation systems, including DC, AC, and stepper motors.

**Course Contents**

Introduction to mechatronics systems. Electrical components of mechatronics; D.C and AC circuit analysis, Semi-conductors, Measurement systems: Sensor and transducers: performance terminology, Sensor Classifications: sensors for displacement, position and proximity; velocity, motion, force, fluid pressure, liquid level. Temperature, light sensors: selection of sensors. Signal conditioning Operations: Amplification, filtering, protection.

Actuation System: Pneumatic and hydraulic systems: directional control valves – rotary actuators. Mechanical actuation systems: cams, gear trains, ratchet and pawl, belt and chai drives, bearings. Electrical actuation systems: mechanical switches, solid state switches, solenoids, dc motors, ac motors, stepper motors.

**ENG 214 Fluid Mechanics (3 Units)**

**Learning Outcomes**

1. Explain the fundamental properties of fluids and evaluate conditions of fluid statics.

2. Apply basic conservation laws to solve practical fluid flow problems.

3. Calculate friction effects and head losses in laminar and turbulent pipe flows.

4. Perform dimensional analysis and utilize dynamic similitude for fluid modeling.

Analyze the principles of construction and operation for selected hydraulic machinery and hydropower systems.

**Course Contents**

Properties of fluids, Fluids statics, Basic conservation laws, friction effects and losses in laminar and turbulent flows in ducts and pipes. Dimensional analysis and dynamic similitude, principles of construction and operation of selected hydraulic machinery. Hydropower systems.

**ENG 222 Basic Electrical Engineering II (3 Units)**

**Learning Outcomes**

1. Explain the atomic structure and basic principles of semiconductors, including intrinsic and extrinsic materials, charge carriers, and energy bands.
2. Describe and calculate key semiconductor properties such as effective mass, carrier mobility, conductivity, carrier lifetime, and recombination.
3. Apply carrier transport equations and analyze semiconductor behavior using continuity equations, drift and diffusion models, and the Hall effect.
4. Analyze the structure and function of PN junction diodes, and evaluate their applications in rectification, filtering, and regulation circuits.
5. Explain and compare the operation of bipolar junction transistors (BJT) in CB, CE, and CC configurations, including input/output characteristics.
6. Design and interpret transistor-based circuits such as small-signal amplifiers and switching devices using AC/DC load line techniques.

**Course Contents**

Introduction to Electrical Machines – DC Motors, DC Generators, AC Motors, AC Generators, Transformers. Three phase balanced circuits, PN junction Diode, Transistors, Thyristors FETs, Zener, Rectifiers. Basic control systems, open/closed loop systems. Fundamentals of communications, introduction of TV, Radio, Telephone systems.

**ENG 224 Computer Programming and Applications (2 Units)**

**Learning Outcomes**

**1.** Design program logic using pseudo-code and flowcharts.

2. Write and execute basic structured programs using languages such as QBASIC, C/C++, and FORTRAN.

3. Develop introductory-level applications using visual programming (Visual Basic) and object-oriented programming (Java).

4. Apply programming logic to model and solve engineering problems.

5. Debug and interpret program outputs across multiple programming environments.

**Course Contents**

Program design using pseudo-code/flow chart. Introduction to Digital Computer Programming, Elements of Structured Basic such as QBASIC, Elements of C/C++, Elements

of FORTRAN, Elements of Visual Basic and Elements of Java. Extensive examples and exercises in solving engineering problems using the programming languages.

**ENG 232 Engineering Drawing II (2 Units)**

**Learning Outcomes**

1. Interpret intersecting solids and generate accurate developments of surfaces.

2. Create detailed manufacturing drawings of cams, belts, chains, and gears.

3. Illustrate mechanical arrangements including bearings, lubrication systems, couplings, brakes, and universal joints.

4. Compile complete assembly drawings from detailed component parts.

5. Review and revise existing engineering drawings to meet standard specifications.

**Course Contents**

Cams. Interpretation of solids. Development of surfaces. Detail drawing. Belts, Chains, Gears. Bearing and lubrication arrangements. Couplings brakes, Flexible shafts, Universal joints, etc. Assembly drawings. Revisions.

**ENG 234 Engineering Mechanics II (2 Units)**

**Learning Outcomes**

1. Apply Hooke's law to determine stresses and strains resulting from mechanical loading and thermal variations.

2. Analyze structural members under torsion and combined loading using analytical methods and the stress circle.

3. Calculate the deflection of beams subjected to symmetrical and combined loads.

4. Evaluate the structural integrity of columns by determining their elastic buckling points.

5. Construct accurate shear force and bending moment diagrams using structural analysis techniques.

**Course Contents**

Hooke's law: stresses and strain due to loading and temperature changes. Torsion. Stress circle. Deflection of beams with symmetrical and combined loadings. Elastic buckling of columns. Shear forces and bending moments. Analytical methods for structures.

**ENG 262 Workshop Technology II (2 Units)**

**Learning Outcomes**

1. Perform advanced machining operations on turret lathes, universal milling machines, and specialized presses.

2. Design and utilize simple jigs and fixtures for the mass production of components.

3. Apply standard metal joining techniques including arc (MIG, MAG, TIG) and gas welding.

4. Conduct and interpret destructive and non-destructive testing (e.g., tensile, ultrasonic, X-ray, dye testing) for production inspection.

5. Execute and document basic laboratory experiments in thermodynamics, applied mechanics, fluid mechanics, and applied electricity.

**Course Contents**

Laboratory investigation and report submission for selected experiments and projects in Thermodynamics, Applied Mechanics and Applied Electricity and Fundamentals of Fluid Mechanics.

i. Machining: (24 Hours) – More advanced work in turning, drilling and milling operations. Use of shaping machines, presses: fly press, hydraulic press. Introduction to jigs and fixtures for mass production of components.
ii. Fabrication: (18 Hours) – Introduction to metal joining techniques: bolting, riveting and welding (arc and gas). Welding machine and equipment: Arc welding (MIG, MAG and TIG). Gas welding and cutting. Chemical bonding.
iii. Production Inspection: Production inspection techniques (destructive and non-destructive); tensile testing, fatigue testing, crush testing, drop testing. Ultrasonic testing, x-ray testing, magnetic particle testing, dye testing, optical scanning test.
iv. Electrical Wiring System: Electrical wiring system: domestic and industrial.
v. Civil Engineering Laboratories: Civil Engineering basic laboratory work.
vi. Electrical / Electronic Engineering Laboratories: Electrical / Electronic Engineering basic laboratory work.
vii. Mechanical Engineering Laboratories: Mechanical Engineering basic laboratory work.

**ENG 282 Engineering Mathematics II (3 Units)**

**Learning Outcomes**

1. Solve second-order differential equations relevant to engineering systems.

2. Evaluate line, multiple, and differentiated integrals and apply them to physical geometries.

3. Analyze analytical functions of complex variables and apply transformation and mapping techniques.

4. Utilize special mathematical functions to solve advanced engineering problems.

5. Implement programmatic solutions to advanced mathematical models using MATLAB, MathCAD, or Excel.

**Course Contents**

Second order differential equations, line integral, multiple integral and their applications, differentiation of integral. Analytical functions of complex variables. Transformation and mapping. special functions. Applications of Microsoft Excel, MATLAB and/or MathCAD.

**ENG 284 Engineer-in-Society (2 Units)**

**Learning Outcomes**

1. Trace the history of engineering and technology and discuss the philosophy of science.

2. Identify the roles of professional engineering bodies and apply the engineers' code of conduct and ethics.

3. Analyze the engineer's role in nation-building, business, politics, and risk management.

4. Evaluate the environmental effects of industrial materials, chemicals, plastics, and radiochemical hazards.

5. Discuss the management of renewable and non-renewable energy resources within the context of global sustainability.

**Course Contents**

Philosophy of Science and Engineering. History of Engineering and Technology. The Engineering profession - engineering - engineering literacy professional bodies and engineering societies. Engineers' code of conduct and ethics. Engineers and nation building - economy, politics, business, safety in Engineering and introduction in Risk analysis, invited lecturers from professionals.

Man – his origin and nature; Man and his cosmic environment; Scientific methodology, Science and technology in the society and service of man. Renewable and non-renewable resources – man and his energy resources. Environmental effects of chemical plastics, Textiles, Wastes and other materials, Chemical and radiochemical hazards, Introduction to the various areas of science and technology. Elements of environmental studies.

**GST 212 Introduction to Entrepreneurship (2 Units)**

Introductory Entrepreneurial skills: Relevant Concepts: Enterprise, Entrepreneur, Entrepreneurship, Business, Innovation, Creativity, Enterprising and Entrepreneurial Attitude and Behaviour. History of Entrepreneurship in Nigeria. Rationale for Entrepreneurship, Creativity and Innovation for Entrepreneurs. Leadership and Entrepreneurial Skills for coping with challenge. Unit Operations and Time Management. Creativity and Innovation for Self-Employment in Nigeria. Overcoming Job Creation Challenges. Opportunities for Entrepreneurship, Forms of Businesses, Staffing, Marketing and the New Enterprise. Feasibility Studies and Starting a New Business. Determining Capital Requirement and Raising Capital. Financial Planning and Management. Legal Issues, Insurance and Environmental Considerations. Also to be incorporated, on the other side of the spectrum, are employability skills – interview techniques, Oral Presentation Skills, etc.

**AFE 202 Food Production and Health Awareness (1 Unit)**

Concept of Food Security: Fundamentals of Food, Food Security, Food Insecurity, Consequences of Food Insecurity, Origin of Food Crisis in Nigeria Agriculture. Ways to Improve Food Security. Methods of food Preservation. Arable Crops Production. Animal Production in the Tropics. Direct Role of Livestock to Food Security. Poultry Production: Housing System in Poultry. Health Determinants, Personal Health Practices and Coping Skills, Healthy Child Development, Health Services Available.

**300 LEVEL FIRST SEMESTER**

**EEE 313: Electronics I (Physical)**

**Learning Outcomes**

1. Explain free electron motion in static electric and magnetic fields.

2. Apply the theory of energy bands to distinguish between conductors, insulators, and semiconductors.

3. Analyze charge carrier transport phenomena and conductivity in crystalline solids.

4. Evaluate the electrical characteristics of discrete devices including diodes, BJTs, photocells, and LEDs.

5. Describe elementary fabrication techniques and basic Integrated Circuit (IC) technology.

**Course Contents**

Free electron motion in static electric and magnetic fields, electronic structure of matter, conductivity in crystalline solids. Theory of energy hands in conductors, insulators and semi-conductors: electrons in metals and electron emissions; carriers and transport phenomena in semi-conductors, characteristics of some electron and resistors, diodes, transistors, photo cell and light emitting diode. Elementary discrete devices fabrication techniques and IC technology.

**EEE 315: Electromagnetic Field and Wave Theory I**
**Learning Outcomes**

1. Understand the fundamental mathematical tools such as vector analysis, calculus, integra theorems and dimensional analysis
2. Understand the behaviour of electric and potential around different structures; surfaces and volumes.
3. Analyze the fundamental laws governing electric and magnetic fields.
4. Apply the concept of Maxwell's Equations and their role in analyzing electromagnetic fields and waves.
5. Compare the behavior of wave propagation mechanism in different media and their effect.
6. Justify the concept of electromagnetic application in the modern society.

**Course Content**

Review of electromagnetic laws, Gauss' law, Ampere's and Faraday's laws, Electric potential, Electrostatic fields due to distribution of charges, Energy stored in Electric field, magnetic fields in and around current carrying conductors of different shapes, magnetic vector potential. Time varying magnetic and electric fields, conduction and displacement current in magnetic vector potential.

**EEE 317: Electrical/Electronics Laboratory 1**

**Course Contents**

Laboratory work involves soldering techniques, measuring instrument, identification of semiconductors, application of circuit and transformer testing. A Laboratory work designed to illustrate topics covered in measurement and instrumentation, electric circuits theory transformers and DC machines.

**EEE 319: Measurements and Instrumentation**

**Learning Outcomes**

1. Identify and understand the various electrical quantities, units, and symbols

2. Understand the theory of errors, the different types of errors and the means of overcoming the errors.

3. Understand the A.C and D.C measurements

4. Know the features and calculations involved in using various indicating instruments and oscilloscope

**5.** Understand the concept of digital test instruments

**Course Contents**

Units. Theory of Errors. Direct and Indirect Measurements. General Instrumentation, Moving Coil Instruments (PMMC). Moving Iron Instruments. Basic Meter in DC measurement. Basic meter in AC measurements; rectifier voltmeter, electro-dynamometer and Wattmeter, instrument transformers; DC and AC bridges and their applications; general form of AC bridge universal impedance bridge; Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital voltmeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO, Instruments for generating and analyzing waveforms; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes; Analog and digital data acquisition systems: tape recorders, D/A and A/D conversions, sample and hold circuits.

**EEE 321: Electric Circuit Theory I**

**Learning Outcomes**

1. Analyze the characteristics and applications of various circuit elements using fundamental circuit theorems

2. Evaluate the transient and steady-state network responses to step, ramp, and impulse inputs in electrical circuits

3. Assess the behavior of network functions and their responses to exponential and sinusoidal sources in dynamic systems

4. Design and synthesize electrical networks using Laplace transform, pole-zero analysis, and resonance techniques, including ladder and two-point configurations

5. Apply Star-Delta transformation and T/Pi network configurations to optimize circuit performance and analyze complex network topologies

**Course Contents**

Circuit elements, circuit theorems, applications. Network response to steps, ramp, impulses. Network functions, response to exponential, sinusoidal sources. Laplace transform, polezero analysis, network synthesis, resonance, two-point analysis, ladder network. Star-Delta transformation, T, P Networks.

**EEE 325: Electrical Machines I**

**Learning Outcomes**

1. Explain the basic principles of relays, actuators, and electromechanical energy conversion.

2. Describe the operation, performance parameters, and equivalent circuits of transformers.

3. Apply concepts of efficiency, regulation, and per-unit values in transformer analysis.

4. Differentiate between types of transformers, including autotransformers and instrument transformers, and discuss their design elements.

5. Analyze transformer behavior in polyphase circuits and conditions for parallel operation.

6. Examine the construction, excitation methods, armature reaction, and performance characteristics of DC generators and motors.

7. Evaluate and select suitable motors and generators for practical applications, including speed control, electric braking, cross-field, and commutator machines.

**Course Contents**

Basic principles of relays and actuators. Transformation of Electric energy. Transformer performance; equivalent circuits, efficiency, regulation, per unit values. Types of Transformers: Auto transformer, Instrument Transformer, Elements of Transformer design. Transformer in Polyphase circuits. Parallel operation of transformers. Basic principles of electromechanical energy conversion. Direct current machines: armature windings, internal torque, and methods of excitation. Armature reaction. Characteristics of D.C. generators and Motors. Basic principles of selection of motors and generators for practical application. Speed control and electric braking. Cross-field machines. Commutator machines.

**EEE 341: Software Development Techniques**

**Learning Outcomes**

1. Apply the software development life cycle and top-down design methodology using pseudo-code and flowcharts.

2. Write structured programs in C utilizing arrays, strings, dynamic memory, and pointer manipulation.

3. Design programs that utilize advanced data structures, including structures, unions, linked lists, and stacks.

4. Implement standard file handling functions for binary and random-access files. Develop, compile, and debug C software in modern environments using static/dynamic libraries and makefiles.

**Course Contents**

Software development life cycle. Top-Down design. Program, design using pseudo-code, flowchart. Flowchart ANSI symbols and usage. Extensive examples, and exercises using pseudo-code/flowchart to solve practical problems in engineering. Debugging and documentation techniques. Programming using a structural language such as C: Symbols, keywords, identifiers, data types, operators, various statements, operator precedence, type conversion, conditional and control structures, function, recursive functions. Arrays: 1-D, and multi-dimensional arrays, passing elements or whole array to a function. Simple sorting and searching on arrays, pointers, strings, dynamic memory allocation. Structures and Unions: Structure declaration and definition, accessing structures, array of structures, pointers and structures, union declaration, enumerated variables. File Handling: Concept of a file, files and streams, standard file handling functions, binary files, random access files. Advanced Topics: Command line parameters, pointers to functions, creation of header files, stacks, linked lists, bitwise manipulation. Software development in C in MS Windows, UNIX/LINUX environments, header file, preprocessor directives, make, makefile. Static and dynamic linking libraries. Extensive examples, and exercises programming in C to solve practical problems in engineering. Exercises are to be done in the Computer Laboratory.

**ENG 381 Engineering Mathematics III (3 Units)**

**Learning Outcomes**

1. Solve systems of linear equations using matrix algebra, determinants, inverse matrices, and Eigen theory.

2. Perform coordinate transformations across polar, cylindrical, and spherical geometries.

3. Evaluate ordinary, double, triple, line, and surface integrals for functions of several variables.

4. Apply vector calculus (gradient, divergence, curl) and integral theorems (Gauss, Green, Stokes) to spatial fields.

5. Analyze singular valued and analytical functions using Cauchy-Riemann equations **and contour integration.**

**Course Contents**

Linear Algebra. Elements of Matrices, Determinants, Inverses of Matrices, Theory of Linear Equations, Eigen Values and Eigen Vectors. Analytical Geometry, Coordinate Transformation, Solid Geometry, Polar, Cylindrical and Spherical Coordinates. Elements of Functions of Several Variables, Surface Variables. Ordinary Integrals, Evaluation of Double Integrals, Triple Integrals, Line Integrals and Surface Integrals. Derivation and Integrals of Vectors, The Gradient of Scalar quantities. Flux of Vectors, The Curl of a Vector Field, Gauss, Greens and Stoke's Theorems and Applications. Singular Valued Functions. Multivalued Functions, Analytical Functions, Cauchy Riemann's Equations. Singularities and Zeroes, Contour Integration including the use of Cauchy's Integral Theorems, Bilinear Transformation. Applications of Microsoft Excel, MATLAB and/or MathCAD.

**AFE 301 Practical Entrepreneurship Skills (1 Unit)**
Profiles of business ventures in the various business sectors such as:

Soap/Detergent, Tooth brush and Tooth paste making; Photography; Brick making; Rope making; Brewing; Glassware production/ Ceramic production, Paper production; Water treatment/conditioning/packaging; Food processing/preservation/packaging; Metal fabrication; Tanning industry; Vegetable oil extraction; Farming; Fisheries/aquaculture; Plastic making; Refrigeration/Air-conditioning; Carving, Weaving; Bakery; Tailoring; Printing; Carpentry; Interior Decoration; Animal husbandry etc. Case Study Methodology applied to the development and administration of Cases that bring out key issues of business environment, start-up, pains and gains of growth of businesses, etc. with particular reference to Nigerian businesses. Experience sharing by business actors in the economy with students during Case presentations.

**300 LEVEL SECOND SEMESTER**

**EEE 314: Electronics II (Analogue)**

**Learning Outcomes**

1. Analyze and design broadband, narrowband, and power amplifiers, as well as voltage/current stabilizing circuits.

2. Design sinusoidal (RC, RLC, crystal) oscillators and non-sinusoidal wave-shaping multi-vibrators.

3. Apply operational amplifiers and timer ICs to analog circuit design.

4. Analyze and interface various digital logic gate families (RTL, TTL, MOS) and flip-flops, counters, and registers.

5. Explain the foundational principles of Digital-to-Analog (D/A) and Analog-to-Digital (A/D) conversion.

**Course Contents**

Broadband and narrow band amplifiers, power amplifiers, voltage and current stabilizing circuits. Sinusoidal RC and RLC and crystal oscillators. Switching properties of electronic devices. Switching and wave-shaping circuits. Generation of non-sinusoidal waveforms, astable, monostable and bistable multi-vibrators, comparator, Schmitt-trigger and time base generators using discrete transistor, operational amplifier fundamentals, analyses and applications, other analog integrated circuits-Time chips and their applications. Analysis and design of logic gates of various families. Diode logic, RTL, TTL, RCL, MOS and MOS of digital integrated circuits. Interfacing between various logic families. Concepts of small, medium, large and very large integration and their consequences, some digital building blocks, flip-flops, counters, register and decoders. Introduction to D/A and A/D conversion principles.

**EEE 316: Electromagnetic Field and Wave Theory II**

**Learning Outcomes**

1. Derive and interpret Maxwell's equations in both differential and integral forms using vector calculus across various coordinate systems.
2. Apply scalar and vector potential formulations to solve electromagnetic field problems and calculate associated potentials.
3. Analyze electromagnetic wave properties such as phase velocity, polarization, energy flow, and wave behavior at media interfaces using boundary conditions and the Poynting vector
4. Evaluate wave propagation in conductors and ionized media, including phenomena like skin effect and plasma frequency, and assess their impact on signal transmission.
5. Design and analyze transmission lines, waveguides, and antennas, calculating key parameters and applying electromagnetic theory to practical communication systems.

**Course Content**

Maxwell's equations (in regular coordinates and vector-calculus notation); Derivation of Maxwell's equation, electromagnetic potential and waves, pointing vector, Boundary conditions, wave propagation in good conductors, skin effect, plane waves across boundaries of different media, propagation of electromagnetic waves in ionized media, fundamentals of transmission lines, matching, voltage reflection coefficient standing wave ratio waveguides and antennae. Applications of wave theory in communication.

**EEE 318: Electrical/Electronic Laboratory II**

Laboratory work involves winding of Transformer and Protective devices. Confirmation of semiconductor-Op Amp, Timing circuit. A Laboratory work designed to illustrate topics covered in electric circuits theory electronic engineering and AC machines.

**EEE 322: Electric Circuit Theory II**

**Learning Outcomes**

1. Apply node and loop analysis techniques to solve complex electrical networks and validate circuit behavior under various conditions.

2. Design and realize transfer functions for electrical systems, including the synthesis and implementation of active networks using Foster and Cauer forms.

3. Analyze and interpret the characteristics and performance of analog filters such as low-pass, high-pass, band-pass, Butterworth, and Chebyshev filters.

4. Evaluate non-linear circuit responses using locus diagrams and assess the impact of non-linearity on system performance.

5. Assess and synthesize network functions for both passive and active systems, demonstrating proficiency in filter design and network optimization.

**Course Outline**

Networks; Node, loop analysis, Realization of transfer functions, Filters: design, operation, low, high, Band pass. Butterworth, Chebychev, filters, Non-linear circuit analysis Locus diagrams. Network functions. Active network synthesis and analysis: Foster and Cauers

**EEE 324: Linear Systems**

**Learning Outcomes**

1. Develop mathematical models of electrical, mechanical, and thermal systems using system analogies and differential equations.

2. Explain the classifications of control systems, and describe feedback concepts including their impact on system performance.

3. Analyze the transient and frequency responses of linear systems using time-domain and frequency-domain methods

4. Assess the stability of control systems using techniques such as Routh-Hurwitz, Bode, and Nyquist criteria

5. Construct and simplify transfer functions, block diagrams, and signal flow graphs to describe and analyze control systems

**Course Contents**

Mathematical models of physical system. Analogues concepts in electrical mechanical and thermal systems. Transfer functions. Block diagrams and signal flow graphs. Feedback control system; advantages, transient and steady-state response of systems. Pole stability. The root-locus methods. Frequency response of systems; Bode and polar plots system stability. Routh and Nyquist criteria. Introduction to analogue computer simulation.

**EEE 326: Electrical Machines II**

**Learning Outcomes**

1. Explain the principle of operation of A.C. machines, including the production of rotating magnetic fields in three-phase systems.

2. Analyze the construction and working principles of three-phase induction motors

3. Compare and contrast the performance and applications of single-phase and three-phase motors.

4. Describe the basic operating principles and applications of synchronous machines.

**5.** Apply knowledge of A.C. motor characteristics to select appropriate motors for practical applications.

**Course Contents**

Magnetic circuits. Basic principles of relays and activators; Ideal transformer. Equivalent circuits and basic analysis of practical transformers. D.C. machine contraction, characteristics of D.C. generators. Excitation of D.C. machines. Torque-speed characteristics of D.C. motors. A.C. machines; production of rotating magnetic fields. Simple theory of three-phase induction motors; torque speed characteristics, three-phase induction motors Single-phase motors-applications. Selection of motors, for practical applications. Synchronous machines.

**EEE 328: Communication Principles**

**Learning Outcomes**

1. Describe the general features and component characteristics of point-to-point, broadcasting, radar, and telemetry communication systems.

2. Analyze and compare analog modulation and demodulation methods, including AM (DSB, SSB, VSB) and FM/PM (wide/narrow band).

3. Evaluate error-control coding, multiplexing, and spread-spectrum signaling in the presence of noise.

4. Analyze the properties and attenuation of various transmission media (open space, cable, fiber optics).

5. Apply the sampling theorem to pulse modulation techniques (PAM, PWM, PCM, DM) and assess error correction mechanisms.

**Course Contents**

Introduction to communication systems; General features of point-to-point communication systems over the entire frequency, characteristics of transmitters, receivers and antennae. Radio and TV broadcasting. Telephony. Facsimile. Radar Telemetry. Analogue modulation systems. Amplitude modulation and demodulation methods, DSB, DSBSC, SSB, and VSB. Comparison of AM systems. Angle modulation and demodulation. Frequency and Phase modulation. Wide band and narrow band and FM.

Study of communication systems design and analysis. Topics include signals and spectra, error-control coding, synchronization, multiple-access techniques, spread spectrum signaling, fading channels baseband signaling and detection in noise, digital and analog modulation and demodulation techniques, and communications link budget analysis.

Transmission media; attenuation in open space, air, cable and fibre channels; construction of cables and fibres, sampling theorem, pulse amplitude modulation, pulse width modulation, multiplexing, quantization systems and pulse code modulation, delta modulation, courses and correction of errors in PCM and DM.

**ENG 382 Engineering Mathematics IV (3 Units)**

**Learning Outcomes**

1. Obtain series solutions for second-order linear differential equations, including Bessel and Legendre equations.

2. Solve two- and three-dimensional boundary value problems (e.g., Sturm-Liouville) using the separation of variables.

3. Apply integral transforms (Fourier, Laplace, Mellin, Hankel) and their inverses to solve engineering systems.

4. Utilize the calculus of finite differences and interpolation formulae for data analysis.

5. Implement numerical integration, differentiation, and ODE/PDE solvers (e.g., Runge-Kutta) utilizing computational software.

**Course Contents**

Series solution of second order linear differential equations with variable coefficients. Bessel and Legendre equations. Equations with variable coefficients. Sturn-Louville boundary value problems. Solutions of equations in two and three dimensions by separation of variables. Eigen value problems. Use of operations in the solution of partial differential equations and Linear integral equations. Integral transforms and their inverse including Fourier, Laplace, Mellin and Handel Transforms. Convolution integrals and Hilbert Transforms. Calculus of finite differences. Interpolation formulae. Finite difference equations. Runge-Kutta and other methods in the solutions of ODE and PDEs. Numerical integration and differentiation. Applications of Microsoft Excel, MATLAB and/or MathCAD.

**ENG 384 Engineering Economics (2 Units)**

**Learning Outcomes**

1. Perform economic analysis to make capital investment decisions and evaluate engineering project alternatives.

2. Apply standardized valuation methods and appraisal standards to real property, equipment, and industrial assets.

3. Evaluate depreciation, obsolescence, and risk within corporate financial practices.

4. Estimate basic capital and operating costs for engineering projects.

5. Develop project plans and schedules, differentiating the roles of the cost engineer during design and construction phases.

**Course Contents**

Economic analysis of engineering projects; value systems economic decisions on capital investments and choice of engineering alternatives; new projects, replacement and abandonment policies, risky decisions; corporate financial practices.

Objectives of valuation work/valuer's primary duty and responsibility. Valuer's obligation to his or her client, to other valuers, and to the society. Valuation methods and practices. Valuation reports. Expert witnessing. Ethics in valuation. Valuation I Appraisal standards. Price, cost and value. Depreciation and obsolescence. Valuation terminology. Appraisal reporting and review. Real property valuation. Personal property valuation. Machinery and equipment valuation. Oil and gas valuation. Mines and quarries valuation. Cost and schedule management- an engineering function. Supporting skills and knowledge. Role of cost engineer during evaluation phase. Role of cost engineer during the basic design phase. Role of cost engineer in contractor selection. Role of cost engineer during detailed engineering design phase. Role of cost engineer during construction. Cost engineering

function as distinct from Design engineering function. Canon of ethics for cost engineers. Basic capital cost estimating. Basic operating cost estimating. Basic project planning and scheduling. Cost engineering terminology. Cost engineering standards.

**400 LEVEL FIRST SEMESTER**

**EEE 405: Numerical Computations**

**Learning Outcomes**

1. Outline the fundamental methods of analyzing errors in numerical systems

2. Propose the convergence theory of computational solution to engineering problems

3. Design numerical algorithms using numerical software systems

4. Compare various numerical methods such as Newton's method, numerical computation of Eigen values and Eigen vectors, Simpson's rules and Euler's method

5. Analyze the concept of numerical integration and differentiation

6. Develop various models using MATLAB for numerical computation and engineering applications

**Course Contents**

Computer as a number crunching tool. Floating-point number representation and arithmetic error, stability, convergence theory of computational solution to problem: numerical algorithm formulation and design, numerical software systems. Iterative solutions of non-linear systems: (Newton's Method). Numerical solutions of linear systems. Numerical computation of Eigen values and Eigen vectors. Curve fitting; function approximation. Numerical differentiation and integration (Simpson's Rule etc). Explicit and implicit methods. Differential equation (Euler's method etc). The Eigen value problem and solution of ordinary differential equations-methods (ODE) of Taylor, Euler predictor-corrector and Runge-kuta. Numerical integration and differentiation. Linear algebra. Finite differences. Introduction to the use of Octave, Matlab or LabView in numerical computation and engineering applications. Emphasis is on the use of software to solve real problems.

**EEE 415: Assembly Language Programming**

**Learning Outcomes**

1. Explain the architecture, instruction formats, and addressing modes of modern microprocessors (Intel 80x86/Motorola 680x0).

2. Write, hand-assemble, and debug assembly language programs utilizing arithmetic, logical, and control instructions.

3. Implement modular programming and write interrupt service routines.

4. Interface low-level assembly language code with high-level languages like C.

5. Program floating-point unit operations and introduce MMX/SSE programming for optimized engineering solutions.

**Course Contents**

Introduction: Language level of abstraction and effect on machine, characteristics of machine code, advantages, justifications of machine code programming, instruction set and dependency on underlying processor. Intel 8086 microprocessor assembly language programming: Programming model as resources available to programmer, addressing modes, instruction format, instruction set-arithmetic, logical, string, branching, program control, machine control, input/output , etc; assembler directives, hand-assembling, additional 80x86/Pentium instructions. Modular programming. Interrupt and service routine. Interfacing of assembly language to C . Intel 80x87 floating point programming. Introduction to MMX and SSE programming. Motorola 680x0 assembly language programming. Extensive practical engineering problems solving in assembly language using MASM for Intel, and cross-assembler for Motorola.

**EEE 423: Electrical/Electronic Laboratory III**

Laboratory work involves winding techniques. Computer hardware and software. Maintenance and repairs, communication. A laboratory work set up to illustrate topics covered in control theory, communication principles, electronic engineering, computer techniques and electric power principles.

**EEE 427: Data Communication and Network**

**Learning Outcomes**

1. Explain fundamental concepts of data transmission and synchronization techniques in communication systems.

2. Analyze error detection and correction methods, including block codes, CRC, and data compression algorithms.

3. Evaluate the structure and function of network protocols, including the OSI model and IEEE 802 standards.

4. Design and simulate basic LAN architectures using medium access control techniques and client-server models.

5. Assess the role of network operating systems and Internet protocols in modern communication systems, including IPv4/IPv6 and system security.

**Course Contents**

Introduction to Data communications: the Development of Data Communications; types and sources of data, simple communications network, transmission definitions, one way transmission, half duplex transmission, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronization, character synchronization, character synchronization, synchronous transmission, asynchronous transmission, efficiency of transmission, error detection methods and data compression. Protocols: Introduction to network protocol. Seven Layer ISO-OSI standard protocols and network architecture. Transport protocols, session services protocols, and other protocols. Institute of Electrical and Electronics Engineering 802 standards. Error control and Data Compression: Forward Error Control; error detection methods; parity checking; linear block codes, cyclic redundancy checking; feedback error control, data compression, Huffman coding and dynamic Huffman coding. Local Area Networks: medium access control techniques – Ethernet, token bus and token ring; LAN standards; fibre distributed data interface, metropolitan area network. Peer-to-peer, Client Server. Client-Server Requirements: GUI design standards, interface independence, platform independence, transaction processing, and connectivity, reliability, backup and recovery mechanisms. Information Network Software; Features and benefits of major recovery mechanisms. Information Network Software: features and benefits of major Network Operating Systems. Network OS: (e.g. Novell NetWare, UNIX/LINUX, OS/2 & Windows NT). TCP/IP and Network OS. INTERNET: Definition, architecture, services, Internet addressing. Internet protocol, IPv4, IPv6. Internet programming, Intranet. System administration, and security issues.

**EEE 431: Electric Power Principles**

**Learning Outcomes**

1. Represent the structure of power systems and explain different types of power station

2. Assess the economic operation of power systems under various operating conditions

3. Analyze power factor correction

4. Explain the polyphase theory and evaluate the voltage profile in D.C. and A.C. distribution circuits

5. Describe different types of material for overhead transmission lines and solve inductance and capacitance of transmission lines

6. Explain the concept of skin effect, corona effect, bundled conductors

7. Describe different types of load forecasting methods

**Course Contents**

Types of power station, operation, auxilliaries, economics of operation - stations, substations power supply economics, tariffs. Power factor correction. Polyphase theory. DC, AC power distribution, network calculations, Overhead line conductors. Corona effect, voltage control, circuit breakers, load forecast, siting of generating plants.

**EEE 441: Control Systems and Servomechanism**

**Learning Outcomes**

1. Explain the fundamentals of feedback control systems and their components.

2. Model dynamic systems (electrical, hydraulic, pneumatic, thermal) using transfer functions.

3. Analyze system response and stability using time-domain and frequency-domain methods (Bode, Nyquist, Root Locus).

4. Design compensators and controllers (PID, Lag, Lead, Lead-Lag) for stability, tracking, and regulation objectives.

5. Demonstrate practical application of servomechanisms (motors, tachogenerators, actuators, error detectors).

**Course Contents**

Review of feedback control systems: Transfer functions and System modelling (electrical, hydraulic, pneumatic and thermal systems). Review of frequency-response analysis (Polar, Bode, Nyquist, Nichols). Review of pole stability and Routh-Hurwitz's stability criterion. System classification and time response analysis. Nyquist stability criterion, M- and N- circles, inverse Nyquist Plots. Bode diagrams and determination of transfer function from asymptotic plots. Relative stability. Introduction to Feedback compensation. The tracking and regulation objectives. Compensation techniques and controller design: Lag, Lead and Lead-Lag compensation; Three-term (PID) controllers (P, PD, PI and PID); Series and parallel compensation design; controller design using Bode, Nichols and Root locus methods. Servomechanical systems: Servo-motors, tachogenerators, error detectors amplifiers, actuators, valves, etc. and their transfer functions. Servomechanism position and velocity control systems. Voltage regulators. Introduction to nonlinear systems.

**EEE 471: Digital Electronics and Systems**

**Learning Outcomes**

1. Explain the basic principles of instrumentation science, including the role of sensors and transducers in measuring physical quantities.

2. Demonstrate the ability to select and utilize appropriate sensors and transducers for various measurement applications, understanding their operational principles and characteristics.

3. Apply the signal processing techniques to filter noise to enhance signal quality in measurement systems and ensure accurate data interpretation

4. Demonstrate proficiency in programming microcontrollers for data acquisition and control applications and integrate hardware with software solutions.

5. Analyse industrial applications of instrumentation, identify challenges and propose effective solutions using learned concepts.

**Course Contents**

Digital Systems and Electronics provides students with a comprehensive understanding of the principles and applications of modern instrumentation, sensors, and electronic systems used in measurement and industrial automation. The course introduces the fundamental concepts of instrumentation science, focusing on the role of sensors and transducers in capturing various physical parameters such as temperature, pressure, force, light, and flow. Students learn signal processing and interfacing techniques to improve data accuracy, reduce noise, and ensure effective communication between hardware and software components. Emphasis is placed on digital logic, microcontroller programming, and system integration, enabling learners to design and implement data acquisition and control systems. The course also explores noise reduction strategies, logic gate operations, and the application of expert system instrumentation, preparing students for real-world challenges in electronics and digital measurement.

**400 LEVEL SECOND SEMESTER**

**ENG 290 Students Work Experience Programme (SWEP) I (3 Units)**
Practices and skills in general engineering through instruction in operation of hand and powered tools for wood and metal cutting and fabrication. Supervised hand on experience in safe usage of tools and machine for selected tasks. General practices on automobile repairs, survey, civil, electrical and other related engineering practice. It is regarded as on the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 200 level).

**ENG 390 Students Work Experience Programme (SWEP) II (3 Units)**
On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 300 level).

**ENG 490 Students Industrial Work Experience Scheme (SIWES) (6 Units)**
On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (24 weeks from the end of the First Semester at 400-Level to the beginning of the First Semester of the following session. Thus, the second semester at 400-Level is spent in industry.)

**500L LEVEL FIRST SEMESTER**

**EEE 511: Digital Signal processing**

**Learning Outcomes**

1. Understand the evolution from analogue to digital systems and the benefits of digital over analogue signal processing.

2. Understand the use of mathematical analytical tools such as z-transform, properties, transfer function, inverse z-transform, z-plane poles and zeros, analysis of linear time-invariant in z-domain for system stability.

3. Analyze the Fourier analysis, discrete Fourier analysis and their various categories.

4. Understand the concept of digital filters, types, and application.

5. Practicalise the application of DSP in audio and video systems.

**Course Contents**

Signal representation in time domain, Fourier transform, Sampling theorem, linear time invariant system, discrete convolution, Z-transform, discrete Fourier transform, discrete filter design. Basic image processing concepts.

**EEE 521: Computer-Aided Design and Simulation**

**Learning Outcome**

1. Operate industry-standard Computer-Aided Design (CAD) software for electronic circuit drafting.

2. Formulate circuit models for simulation using SPICE-based packages (e.g., PSPICE, HSPICE).

3. Simulate and analyze the functional behavior of electronic circuits under varying conditions.

4. Utilize electronic workbench platforms to troubleshoot and optimize circuit designs prior to physical prototyping.

**Course Content**

Computer Aided design & drafting. Simulation of circuit using appropriate packages e.g. PSPICE, HSPICE, Electronic Workbench, Visio technical etc.

**EEE 531: Solid State Electronics**

**Learning Outcomes**

1. Relate the physical and electrical properties of semiconductors (band structure, Fermi functions, doping) to device conductivity.

2. Apply the law of mass action and charge transport equations (drift and diffusion) to semiconductor analysis.

3. Evaluate the I-V characteristics of PN junctions and metal-semiconductor junctions.

4. Analyze the operation principles of Bipolar Junction Transistors (BJTs) and Field Effect Transistors (MOSFETs/C-MOS).

5. Explain the operational physics and applications of optoelectronic and photonic devices (LEDs, Lasers, photovoltaic cells).

**Course Contents**

Fundamental properties of semiconductors and their devices – physical and electrical properties such as band structures; Fermi functions; doping and conductivity. Charge transport properties including drift and diffusion. Presentation of the law of mass action and applications. Junction devices including p-n junction and metal semi-conduction junctions, their I-V characteristics and applications. Other major devices of interest including the bipolar junction transistors (BJTs), metal-oxide field effect transistors (MOSFETs), and C-MOS. Optoelectronic and photonic devices including LED, Lasers, solar (photovoltaic) cells.

**EEE 541: Reliability of Electrical and Electronics System**

**Learning Outcomes**

1. Define the fundamental concepts of reliability, maintainability, and availability in engineering context.

2. Apply elementary reliability theory to evaluate power and electronic systems.

3. Identify types of faults and assess the test characteristics of electrical and electronic components.

4. Design electronic packaging and mounting systems optimized for higher reliability and adequate ventilation.

5. Implement protection strategies to safeguard electronic systems against environmental factors like humidity and dust.

**Course Contents**

Introduction to reliability, maintainability, availability. Elementary reliability theory. Application to power systems and electronics systems. Test characteristics of electrical and electronic components. Types of faults. Design for higher reliability. Packaging, mounting, ventilation. Protection from humidity, dust.

**EEE 551: Power Electronics & Drives**

**Learning Outcomes**

1. Analyze the switching characteristics of semiconductor power devices.

2. Design and evaluate power conversion circuits (AC-DC, DC-DC, DC-AC, AC-AC).

3. Apply Silicon Controlled Rectifiers (SCR) and thyristors for motor control, heating, and lighting applications.

4. Evaluate the use of mechanical relays, solid-state relays, and stepping motors in control architectures.

5. Develop power electronic strategies for regulating and controlling electrical drives.

**Course Contents**

Characteristics of semiconductor switches. Power conversion from AC to DC, DC to DC, DC to AC, AC to AC. Applications of SCR and other Thyristors devices: motor control, control of drives, heating and lighting. Mechanical relays, solid state relays and stepping motors.

**EEE 561: Control Engineering**

**Learning Outcomes**

1. Analyze state-space representations for linear systems and understand their significance in control theory.

2. Demonstrate proficiency in converting between transfer functions and state-space representations and recognize implications for system behavior.

3. Analyse eigenvalues and eigenvectors to assess system stability and response, applying concepts to real-world scenarios.

4. Design and implement state feedback control systems and observers, including compensators and Linear Quadratic Regulators

5. Optimize control strategies in practical applications such as robotics and signal processing using computer-aided tools.

**Course Contents**

Control Engineering provides students with an advanced understanding of modern control theory through the state-space approach, equipping them with both theoretical foundations and practical skills for the analysis and design of dynamic systems. The course begins with the formulation of state-space models and their relationship to transfer functions, enabling learners to bridge classical and modern control representations. Students will explore the role of eigenvalues and eigenvectors in determining system response and stability, while also mastering critical concepts such as controllability and observability that form the backbone of system design. Building on this foundation, the course emphasizes the design and implementation of state feedback controllers, observers, and compensators, with a strong focus on optimal control through the Linear Quadratic Regulator (LQR) framework. Real-world applications, including robotics, circuit synthesis, and signal processing, are integrated to illustrate the industrial relevance of control strategies.

**EEE 571: Student Project I**

Original and individual student project related to a prescribed electrical and electronic engineering problem.

**EEE 535: Microwave & Satellite Communications**

**Learning Outcomes**

1. Explain the principles of wave propagation in waveguides, resonators, and passive components.

2. Analyze the characteristics of microwave passive devices such as couplers, isolators, and circulators.

3. Evaluate the operation of active microwave devices such as klystrons, magnetrons, and traveling-wave tubes.

4. Apply knowledge of semiconductor microwave devices (varactor, PIN, Gunn diodes, photodiodes, phototransistors) to practical circuit design.

5. Demonstrate the role of microwave electronics in modern communication and radar applications.

**Course Contents**

Microwave engineering: Review of interaction between electronics and fields-plane propagation in free space, glossy media and metallic films. Transmission lines and wave guides. Microwave components-cavity resonators, wave guide Tees, directional couplers-circulators and isolators. Microwave devices-tubes, solid state devices. Microwave circuits-impedance transformation and matching, resonant and filter circuits. Radar systems: nature of radar and radar equations, composition of a radar system, application of different types of radar. Introduction to satellite communication: Fundamentals of satellite communication systems. Orbit types, ground stations and support sub-systems. Geostationary and low earth orbit systems and services.

**EEE 537: Antenna Theory and Radar Systems**

**Learning Outcomes**

1. Calculate antenna parameters including radiation patterns, radiation resistance, gain, and effective area.

2. Analyze the design and performance evaluation of traveling wave, small/large HF, and microwave antennas.

3. Explain the fundamental principles of range and direction finding using radar echoes.

4. Evaluate the requirements, limitations, and modulation techniques associated with modern radar systems.

5. Specify and analyze microwave components utilized in radar transmission and reception.

**Course Contents**

Concept of antenna radiation patterns, radiation resistance, gain, effective area, reciprocity. Travelling wave and HF antenna. Analysis, design and evaluation of antenna, small and large HF types. Special microwave antennas. Principle of range and direction finding by means of radio echoes. Requirements and limitation of radar, Modulation and microwave components of radar.

**EEE 553: Power System Engineering I**

**Learning Outcomes**

1. Explain and represent power systems using one-line diagrams, per-unit analysis, and equivalent impedance models.

2. Apply load flow methods to determine bus voltages, power injections, and system operating conditions.

3. Construct and analyze bus admittance and impedance matrices for power system studies.

4. Evaluate short-circuit conditions by calculating fault currents, fault levels, and the impact of protective devices.

5. Compute fault levels for symmetrical faults and design mitigation strategies, such as current limiting reactors, to protect power system equipment.

6. Model sequence networks for unsymmetrical fault analysis and calculate fault currents in different parts of the system.

7. Explain the principles of power system protection, including the operational sequence of fault clearing and the methods used for fault discrimination.

**Course Contents**

Representation of power systems and per-unit quantities. Economic operation of power systems. Power system equation and analysis and load flow studies. Symmetrical Three phase faults on synchronous machines. Bus Impedance matrix in fault calculations. Symmetrical components, symmetrical and unsymmetrical faults. Analysis of unsymmetrical faults using the Bus impedance matrix. Digital calculation of fault currents using the Bus impedance matrix. Digital calculation of fault currents. Elements of power systems stability.

**EEE 555: High Voltage Engineering**

**Learning Outcomes**

1. Analyze and apply operational amplifiers in linear and non-linear circuit design.

2. Design and evaluate advanced analog circuits including, logarithmic amplifiers, multipliers, gyrators and negative impedance converters.

3. Compare and implement ADC/DAC systems in instrumentation

4. Apply 555 timer circuits in various timing applications

5. Demonstrate digital circuit applications in instrumentation

**Course Contents**

Importance of High voltage generation and transmission. Characteristics and details of high voltage equipment with emphasis on line structure and hardware. Generation of high AC/DC and impulse voltages. Requirement of testing of internal and external insulation system. Propagation of surges in high voltage transmission lines. Protection of transmission lines and substation from direct lighting strokes. Preventive testing of insulation process in multilayer conductors and insulation coordination. Line and substation insulation. Over head line, bus-bars, isolators and circuit breakers. Corona/Radio interferences and minimization of their effects on the lines. Brief

discussion about discharge in gases, breakdown voltages in gases. Electric field calculation for different gases, breakdown voltages in gases. Electric field calculation for different electrode configuration with respect to distance, temperature etc.

**EEE 565: Modern Control System**

**Learning Outcomes**

1. Apply state variable modeling, feedback control, and pole placement techniques to dynamic systems.

2. Design asymptotic observers for accurate state measurement and implement combined observer-control compensators.

3. Analyze the stability and transient response of sampled-data and linear discrete-time control systems.

4. Formulate optimal control strategies for dynamic tracking and system regulation.

5. Solve system optimization problems utilizing Linear Quadratic Regulator (LQR) theory.

**Course Contents**

State variable feedback and pole placement. Module controllability. Asymptotic observers for state measurement. Combined observer-control compensators. State-variables and linear discrete-time systems. Analysis of sampled-data control system, stability analysis, compensation techniques. Optimal control: the regulator and tracking problem. The linear quadratic regulator theory.

**500L LEVEL SECOND SEMESTER**

**EEE 512: Digital Communication**

**Learning Outcomes**

1. Explain the ISO-OSI layered architecture, TCP/IP, and X25 protocols in data transmission.

2. Analyze circuit-switched and packet-switched networks, including ARQ error detection and recovery.

3. Apply basic queueing theory and Erlang B/C formulas to evaluate telephone switch blocking probabilities.

4. Design telephone and PABX installations, specifying cables, lighting protection, and earthing techniques.

5. Prepare and evaluate Bills of Engineering Materials for telecommunication installations.

**Course Contents**

The ISO-OSI layered architecture, packet switching and circuit switching error detection and recovery (ARQ) protocols, bridges and routers, basic queueing theory, telephone switches, Erlang-B and Erlang-C blocking formulae TCP/IP, X25. Telephone installations, PABX installations: choice of cables and accessories. Lighting protection and earthing techniques. Bill of engineering material and evaluation and bill of telecommunication installations.

**EEE 522: Microprocessor Application, Organization & Embedded Systems**

**Learning Outcomes**

1. Architect microprocessor organizations and interface memory and peripheral subsystems.

2. Execute hardware-software co-design for complex microprocessor applications.

3. Explain the internal architecture and programming of programmable Digital Signal Processors (DSP).

4. Develop and optimize embedded software for systems requiring real-time performance.

5. Design and implement data scramblers and reliable interfaces for telecommunications hardware.

**Course Contents**

Microprocessor organization and interfacing: memory interfacing. Hardware-software design of microprocessor systems. Introduction to Embedded microcomputer systems. Architecture of programmable digital signal processor. Programming for real-time performance. Design and implementation of data scrambler and interfaces to telecommunications.

**EEE 524: Analog IC design & Applications**

**Learning Outcomes**

1. Analyze and design active filters and signal generators using Analog Integrated Circuits.

2. Implement analog multipliers for signal processing applications.

3. Design precision timing circuits utilizing integrated circuit timers.

4. Specify and apply integrated voltage regulators for reliable power supply design.

5. Explain the operational principles and applications of Phase-Locked Loops (PLL) in circuit synchronization.

**Course Contents**

Analysis and Design of signal generators, active filters. Analogue multipliers and their application. Integrated circuit timers. Integrated voltage regulators. Phase locked loop.

**EEE 552: Industrial Utilization of Electric Power**

**Learning Outcomes**

1. Apply basic laws of illumination to specify and design industrial lighting devices.

2. Explain the mechanisms of electric traction and variable-speed drives, including mechanical and electrical braking.

3. Analyze the operation of cycloconverters, DC regulators, and AC regulators in power applications.

4. Develop four-quadrant control strategies for DC and induction motor thyristor drives.

5. Evaluate the operational methods and industrial applications of electrical induction heating.

**Course Contents**

Illumination: Units and definitions. Basic laws of illumination; illumination devices. Electric traction, mechanical electrical braking variable-speed drives. Four quadrant dc motor drives. Cycloconveters. Dc and Ac regulators. Industrial applications of thyristor drives control. DC motor control. Induction motor control. Induction heating.

**EEE 564: Current Trends in Electrical/Electronic Engineering**

**Learning Outcomes**

1. Identify key software tools used in electrical and electronics engineering for circuit design, simulation, analysis, and control systems.

2. Explain the applications and limitations of specific software tools in solving real-world electrical and electronics engineering problems.

3. Utilize selected software tools to design and simulate a basic electrical or electronic system, such as a circuit or control model.

4. Compare the features and performance of at least two software tools used for a specific electrical engineering task (e.g., circuit simulation or power system analysis).

5. Develop and present a comprehensive seminar on current trends in electrical electronic engineering and the application of a chosen software tool, including a practical demonstration or case study.

**Course Contents**

This course is to address and discuss current development in the field of electrical & electronics engineering. Students are expected to submit a term paper on given topics.

**EEE 554: Power Systems Engineering II**

**Learning Outcomes**

1. Evaluate the performance of transmission lines using Kirchhoff's voltage and current, and Ohm's law, etc.

2. Analyse the power generation and voltage control, and voltage collapses of electricity power grid.

3. Formulate the mathematical equations governing power system stability studies.

4. Interpret the stability of power systems under different disturbances

5. Understand and justify the use of power line carrier systems for high-frequency communication in the power industry

6. Assess the application of deregulation and smart/micro-grid technologies to the Nigerian power grid and the integration of distributed generation

**Course Contents**

Load curves and load forecasting. Control of power generation, frequency and voltage control and reactive power control, voltage collapse, contingency planning, system stability, automatic voltage regulators, regulating transformers. Fault analysis. Protection systems: relays, carrier protection, principles of fault detection, discrimination and clearance in transformers, generators and transmission lines. Review of optimization theory and application to power dispatch problems, generation planning and transmission planning. Electricity deregulation. Introduction to distributed generation and smart grids. Introduction to power system communication; power line carrier & application.

**EEE 566: Introduction to Non-Linear Systems**

**Learning Outcomes**

1. Describe the fundamental characteristics of nonlinear systems and common sources of nonlinearities

2. Formulate and analyze nonlinear differential equations relevant to control systems

3. Apply linearization and approximation methods to simplify and analyze nonlinear systems near equilibrium points

4. Evaluate the behavior of nonlinear control systems through phase-plane analysis, describing function methods, or simulations

5. Develop and test control logic for simple industrial process automation using PLCs

**Course Contents**

Introduction to Nonlinearities and Nonlinear Systems: Non-linear differential equations, characteristics of nonlinear systems, common nonlinearities. Analysis of Nonlinear Systems: Linearization Approximations, Piecewise linear approximation, the Describing Function, Concept and derivation for common nonlinearities, the dual input describing function; stability analysis using the describing function. Limit cycle prediction. Analysis of Nonlinear Systems. The Phase-Plane Method. Construction of phase trajectories, transient analysis by the phase plane method. Lyapunov's Indirect Method. Stability Analysis of Non-linear Systems using Lyapunov's Method. Introduction to Sampled-Data Systems. The z-transforms; Pulse Transfer Functions; Stability Analysis in the z-plane.

**END.**